Convergence, Hypocoercivity, and Mean-Field Limit for the Kinetic Core of the Geometric Gas

Abstract

This paper isolates the pure kinetic backbone of the geometric gas: a confining underdamped Langevin dynamics on $\mathbb{R}^d$, augmented by a row-normalized viscous velocity alignment force and followed, at observation times, by a smooth velocity squashing map. The cloning operator, companion sampling, fitness potential, and state-dependent anisotropic diffusion are deliberately omitted. Within this kinetic-only setting we prove the two structural facts that make the model tractable: the viscous force is uniformly bounded on velocity-capped states, and it is dissipative on relative kinetic energy. We then formulate the finite-particle hypocoercive entropy estimate that yields an N-uniform logarithmic Sobolev inequality and exponential convergence in relative entropy and total variation. Finally we derive the McKean–Vlasov limit for the viscous alignment field, prove quantitative propagation of chaos in relative entropy, and pass the N-uniform logarithmic Sobolev inequality to the mean-field equilibrium. The paper is self-contained in the sense that all assumptions, definitions, and theorems used below are stated explicitly in this document.

Contents

1 Introduction

The purpose of this paper is to extract and organize the convergence theory of the kinetic part of the geometric gas in a form that does not depend on the cloning stage or on the adaptive diffusion geometry. The model studied here is the following.

- The position variable evolves by transport: $\dot{x} = v$.
- The velocity variable is subject to a confining force $-\nabla U(x)$, linear friction $-\gamma v$, isotropic Brownian noise, and a row-normalized velocity alignment force.
- After each observation interval one may apply a smooth radial squashing map ψ to cap the observed speed without destroying the previously obtained probabilistic estimates.

Two themes drive the analysis.

1. The *backbone hypocoercive mechanism* comes from underdamped Langevin dynamics with confinement and friction.
2. The *new viscous term* is harmless at the level of stability because its action is bounded and dissipative on relative velocity disagreements.

The paper therefore proceeds in two layers. First we prove the structural lemmas for the viscous operator and the velocity squashing map. Then we graft those facts onto the hypocoercive framework for the kinetic backbone, obtaining N-uniform entropy and total-variation convergence. The final sections pass to the mean-field equation, prove a quantitative propagation of chaos estimate, and deduce a mean-field logarithmic Sobolev inequality. The analysis uses only tools from Langevin dynamics, hypocoercivity, Harris-type ergodic theory, and McKean–Vlasov limits [5, 4, 7, 6].

2 The Kinetic Geometric Gas Core

2.1 Standing assumptions

Assumption 2.1 (Strongly confining potential)**.** The potential $U : \mathbb{R}^d \to \mathbb{R}$ satisfies:

(i) $U \in C^2(\mathbb{R}^d)$ and there exist constants $0 < m_U \le L_U < \infty$ such that

$$\|\nabla U(x) - \nabla U(y)\| \le L_U \|x - y\| \qquad \forall x, y \in \mathbb{R}^d.$$

(ii) The Hessian is uniformly positive in the sense

$$\langle \xi, \nabla^2 U(x)\xi\rangle \ge m_U \|\xi\|^2 \qquad \forall x, \xi \in \mathbb{R}^d.$$

(iii) U is coercive in the Lyapunov sense:

$$\langle x, \nabla U(x)\rangle \ge \alpha_U \|x\|^2 - \beta_U \qquad \forall x \in \mathbb{R}^d$$

for some $\alpha_U > 0$ and $\beta_U < \infty$.

Remark 2.2. Assumption 2.1(i)–(ii) places the paper in the classical uniformly convex Langevin regime. In this regime the Gibbs measure (7) satisfies a logarithmic Sobolev inequality by the Bakry–Émery criterion, and the modified-entropy computation below yields an explicit hypocoercive decay rate for the kinetic backbone; see [1, 7].

Assumption 2.3 (Regularized Gaussian kernel)**.** Fix $\ell_{\mathrm{visc}} > 0$, $\kappa_0 > 0$, and define

$$K(x, y) := \kappa_0 + \exp\left(-\frac{\|x - y\|^2}{2\ell_{\mathrm{visc}}^2}\right).$$

Then K is symmetric, strictly positive, bounded, and globally Lipschitz in each variable. Set

$$k_* := \inf_{x,y} K(x, y) = \kappa_0, \qquad k^* := \sup_{x,y} K(x, y) = 1 + \kappa_0.$$

Definition 2.4 (Finite-particle viscous weights)**.** For $N \geq 2$ and a configuration $X = (x_1, \ldots, x_N) \in (\mathbb{R}^d)^N$, define

$$d_i(X) := \sum_{k \neq i} K(x_i, x_k), \qquad \omega_{ij}(X) := \frac{K(x_i, x_j)}{d_i(X)} \quad (j \neq i).$$

Then $\omega_{ij} \geq 0$ and $\sum_{j \neq i} \omega_{ij} = 1$.

Definition 2.5 (Finite-N kinetic core)**.** Fix $\gamma > 0$, $\sigma > 0$, and $\nu_{\mathrm{visc}} \geq 0$. For $i = 1, \ldots, N$, the finite-particle kinetic geometric gas core is the Itô SDE

$$dX_i(t) = V_i(t)\, dt, \tag{1}$$

$$dV_i(t) = \Big(-\nabla U(X_i(t)) - \gamma V_i(t) + F_i(X(t), V(t))\Big)\, dt + \sigma\, dW_i(t), \tag{2}$$

where the viscous force is

$$F_i(X, V) := \nu_{\mathrm{visc}} \sum_{j \neq i} \omega_{ij}(X)\big(V_j - V_i\big). \tag{3}$$

The Brownian motions $W_1, \ldots, W_N$ are independent standard d-dimensional Wiener processes.

Definition 2.6 (Mean-field viscous field)**.** Let $f \in \mathcal{P}(\mathbb{R}^{2d})$ with spatial marginal

$$\rho_f(dx) := \int_{\mathbb{R}^d} f(dx, dq).$$

Define the normalized kernel mass

$$m_f(x) := \int_{\mathbb{R}^d} K(x, y)\, \rho_f(dy)$$

and the mean-field viscous field

$$F[f](x, v) := \frac{\nu_{\mathrm{visc}}}{m_f(x)} \iint_{\mathbb{R}^{2d}} K(x, y)(q - v)\, f(dy, dq). \tag{4}$$

The mean-field kinetic equation is

$$\partial_t f_t + v \cdot \nabla_x f_t + \nabla_v \cdot \Big[\big(-\nabla U(x) - \gamma v + F[f_t](x, v)\big) f_t\Big] = \frac{\sigma^2}{2} \Delta_v f_t. \tag{5}$$

Remark 2.7. Because $m_f(x) \geq k_* > 0$ for every probability measure f, Definition 2.6 is globally well posed at the algebraic level: there is no denominator singularity in the mean-field alignment field.

Definition 2.8 (Velocity squashing)**.** Fix $V_{\rm alg} > 0$. The smooth velocity squashing map is

$$\psi(v) := V_{\rm alg} \frac{v}{V_{\rm alg} + \|v\|}, \qquad \psi(0) := 0.$$

The associated observation map on phase space is

$$S(x, v) := (x, \psi(v)).$$

Definition 2.9 (BAOAB observation chain)**.** Fix a time step $\Delta t > 0$. The algorithmic update used at observation times is the BAOAB Strang splitting of the kinetic core:

$$Z_{n+1}^{\rm pre} = \Phi_B^{\Delta t/2} \circ \Phi_A^{\Delta t/2} \circ \Phi_O^{\Delta t} \circ \Phi_A^{\Delta t/2} \circ \Phi_B^{\Delta t/2}(Z_n),$$

where $Z_n = (X_n, V_n)$ and the substeps are the explicit maps

$$\Phi_B^h(X, V) := \big(X,\, V + h\, G(X, V)\big), \qquad \Phi_A^h(X, V) := \big(X + hV,\, V\big).$$

Here $G(X, V) = (G_1(X, V), \ldots, G_N(X, V))$ with

$$G_i(X, V) := -\nabla U(X_i) + F_i(X, V).$$

The Ornstein–Uhlenbeck substep is

$$\Phi_O^h(X, V) := \left(X,\, e^{-\gamma h} V + \sqrt{\frac{\sigma^2}{2\gamma}\big(1 - e^{-2\gamma h}\big)}\,\Xi\right),$$

where $\Xi \sim \mathcal{N}(0, I)$ is an independent standard Gaussian vector. The viscous force F_i is the row-normalized coupling from (3). The observed state is then postprocessed componentwise by

$$Z_{n+1} = S(Z_{n+1}^{\rm pre}) = (X_{n+1}^{\rm pre}, \psi(V_{n+1}^{\rm pre})).$$

This is the same BAOAB ordering used in the implementation and in the convergence-program notes already present elsewhere in the repository.

3 Structural Bounds: Confinement, Viscosity, and Squashing

Lemma 3.1 (Smooth bounded cap)**.** *The map ψ of Definition 2.8 is smooth on $\mathbb{R}^d \setminus \{0\}$, continuous on $\mathbb{R}^d$, satisfies $\|\psi(v)\| \leq V_{\rm alg}$, and is* 1*-Lipschitz:*

$$\|\psi(v) - \psi(w)\| \leq \|v - w\| \qquad \forall v, w \in \mathbb{R}^d.$$

Proof. Write $r = \|v\|$. For $v \neq 0$,

$$\psi(v) = \phi(r) v, \qquad \phi(r) = \frac{V_{\rm alg}}{V_{\rm alg} + r}.$$

Hence

$$D\psi(v) = \phi(r) I + \phi'(r) \frac{v \otimes v}{r} = \frac{V_{\rm alg}}{V_{\rm alg} + r} I - \frac{V_{\rm alg}}{(V_{\rm alg} + r)^2} \frac{v \otimes v}{r}.$$

The radial eigenvalue is $V_{\rm alg}^2/(V_{\rm alg}+r)^2 \le 1$ and each tangential eigenvalue is $V_{\rm alg}/(V_{\rm alg}+r) \le 1$. Therefore

$$\|D\psi(v)\|_{\rm op} \le 1 \qquad (v \neq 0).$$

Any $u \in \mathbb{R}^d$ decomposes as $u = \alpha v/r + u_\perp$ with $u_\perp \perp v$, so

$$\|D\psi(v)u\| \le \frac{V_{\rm alg}^2}{(V_{\rm alg}+r)^2}|\alpha| + \frac{V_{\rm alg}}{V_{\rm alg}+r}\|u_\perp\| \le \|u\|.$$

Hence ψ is 1-Lipschitz on $\mathbb{R}^d \setminus \{0\}$ by the mean-value theorem on line segments. At the origin,

$$\lim_{v\to 0} \|D\psi(v)\|_{\rm op} = 1,$$

so the same Lipschitz bound extends continuously to $v = 0$. The speed cap is obvious from

$$\|\psi(v)\| = \frac{V_{\rm alg}\|v\|}{V_{\rm alg}+\|v\|} \le V_{\rm alg}.$$

□

Lemma 3.2 (Uniform bound on the finite-particle viscous force)**.** *If* $\max_{1\le k\le N}\|V_k\| \le V_*$*, then for every* i

$$\|F_i(X,V)\| \le 2\nu_{\rm visc}V_*.$$

The bound is independent of N*.*

Proof. By Definition 2.4,

$$F_i(X,V) = \nu_{\rm visc}\left(\sum_{j\neq i}\omega_{ij}(X)V_j - V_i\right).$$

Taking norms and using $\sum_{j\neq i}\omega_{ij} = 1$ gives

$$\|F_i(X,V)\| \le \nu_{\rm visc}\left(\sum_{j\neq i}\omega_{ij}(X)\|V_j\| + \|V_i\|\right) \le 2\nu_{\rm visc}V_*. \qquad \square$$

Lemma 3.3 (Dissipativity of the viscous coupling)**.** *Fix the positions* X *and consider only the viscous ODE*

$$\dot V_i = \nu_{\rm visc}\sum_{j\neq i}\omega_{ij}(X)(V_j - V_i).$$

Define

$$D(X) := \sum_{i=1}^N d_i(X), \qquad \bar V_d := \frac{1}{D(X)}\sum_{i=1}^N d_i(X)V_i,$$

and the degree-weighted velocity variance

$$\mathscr{V}_d(V) := \frac{1}{D(X)}\sum_{i=1}^N d_i(X)\|V_i - \bar V_d\|^2.$$

Then

$$\frac{d}{dt}\mathscr{V}_d(V(t)) = -\frac{2\nu_{\rm visc}}{D(X)}\sum_{1\le i<j\le N} K(x_i,x_j)\|V_i - V_j\|^2 \le 0. \tag{6}$$

Proof. Since X is frozen, the degrees $d_i(X)$ are constant in time. Set $\delta_i := V_i - \bar{V}_d$. By symmetry of K,

$$\frac{d}{dt}\bar{V}_d = \frac{\nu_{\text{visc}}}{D(X)} \sum_{i=1}^{N} \sum_{j \neq i} K(x_i, x_j)(V_j - V_i) = 0,$$

so $\dot{\delta}_i = \dot{V}_i$. Therefore

$$\begin{aligned}
\frac{d}{dt}\mathscr{V}_d(V(t)) &= \frac{2}{D(X)} \sum_{i=1}^{N} d_i(X)\langle \delta_i, \dot{V}_i \rangle \\
&= \frac{2\nu_{\text{visc}}}{D(X)} \sum_{i=1}^{N} \sum_{j \neq i} K(x_i, x_j)\langle \delta_i, \delta_j - \delta_i \rangle \\
&= -\frac{\nu_{\text{visc}}}{D(X)} \sum_{i \neq j} K(x_i, x_j)\|\delta_i - \delta_j\|^2 \\
&= -\frac{2\nu_{\text{visc}}}{D(X)} \sum_{1 \leq i < j \leq N} K(x_i, x_j)\|V_i - V_j\|^2,
\end{aligned}$$

which is non-positive. $\square$

Lemma 3.4 (Lipschitz control of the mean-field alignment field)**.** *Let $\mu, \nu \in \mathcal{P}(\mathbb{R}^{2d})$ have finite first moments. There exists a constant $L_{\text{mf}} < \infty$, depending on $\nu_{\text{visc}}, k_*, k^*, \ell_{\text{visc}}$ and on a fixed first-moment bound for the measure class under consideration, such that*

$$\|F[\mu](x, v) - F[\nu](x', v')\| \leq L_{\text{mf}}\big(\|x - x'\| + \|v - v'\| + W_1(\mu, \nu)\big),$$

where W_1 denotes the 1-Wasserstein distance on $\mathbb{R}^{2d}$.

Proof. Write

$$A[\mu](x) := \frac{1}{m_\mu(x)} \iint K(x, y)q\,\mu(dy, dq), \qquad F[\mu](x, v) = \nu_{\text{visc}}\big(A[\mu](x) - v\big).$$

To estimate the denominator, use the coupling characterization of W_1 and the global Lipschitz bound on K:

$$|m_\mu(x) - m_\nu(x')| \leq \iint |K(x, y) - K(x', y')|\,\pi(dy, dq, dy', dq') \leq C_K\big(\|x - x'\| + W_1(\mu, \nu)\big),$$

for any coupling $\pi \in \Pi(\mu, \nu)$ and some constant C_K depending only on L_K. Since $m_\mu, m_\nu \geq k_*$, the reciprocal map satisfies

$$\big|m_\mu(x)^{-1} - m_\nu(x')^{-1}\big| \leq k_*^{-2}|m_\mu(x) - m_\nu(x')|.$$

For the numerator, fix a coupling $\pi \in \Pi(\mu, \nu)$ and write

$$J_\mu(x) := \iint K(x, y)q\,\mu(dy, dq).$$

Then

$$\|J_\mu(x) - J_\nu(x')\| \leq \iint \|K(x, y)q - K(x', y')q'\|\,d\pi.$$

On any class of laws with a fixed first-moment bound in the velocity variable, the integrand is controlled by a constant times $\|x - x'\| + \|y - y'\| + \|q - q'\|$, so

$$\|J_\mu(x) - J_\nu(x')\| \le C_J\big(\|x - x'\| + W_1(\mu, \nu)\big),$$

with C_J depending only on the kernel bounds and the moment bound for the class under consideration. Combining the numerator and denominator estimates, and using

$$A[\mu](x) - A[\nu](x') = \frac{J_\mu(x) - J_\nu(x')}{m_\mu(x)} + J_\nu(x')\big(m_\mu(x)^{-1} - m_\nu(x')^{-1}\big),$$

gives

$$\|A[\mu](x) - A[\nu](x')\| \le C\big(\|x - x'\| + W_1(\mu, \nu)\big).$$

Finally,

$$F[\mu](x, v) - F[\nu](x', v') = \nu_{\text{visc}}\big(A[\mu](x) - A[\nu](x')\big) - \nu_{\text{visc}}(v - v'),$$

so

$$\|F[\mu](x, v) - F[\nu](x', v')\| \le L_{\text{mf}}\big(\|x - x'\| + \|v - v'\| + W_1(\mu, \nu)\big). \qquad \square$$

Proposition 3.5 (Finite- and infinite-dimensional well posedness). *Under Assumptions 2.1 and 2.3,*

(i) *for every $N \ge 2$ the SDE* (1)–(2) *has a unique global strong solution;*

(ii) *for every initial law $f_0 \in \mathcal{P}(\mathbb{R}^{2d})$ with finite second moment, the McKean–Vlasov equation* (5) *has a unique global solution $t \mapsto f_t$ in $C([0, \infty); \mathcal{P}_2(\mathbb{R}^{2d}))$.*

Proof. Fix N and write the drift as

$$b_N(X, V) = \big(V, -\nabla U(X) - \gamma V + F(X, V)\big).$$

Because the kernel K is smooth and strictly positive, every denominator

$$d_i(X) = \sum_{k \ne i} K(x_i, x_k)$$

is bounded below by $(N - 1)k_*$, so each normalized weight $\omega_{ij}(X)$ is smooth in X. Hence b_N is locally Lipschitz on $(\mathbb{R}^{2d})^N$, and standard SDE theory gives a unique strong solution up to the explosion time. To rule out explosion, define

$$\mathcal{E}_N(X, V) := \frac{1}{N}\sum_{i=1}^N \big(\|X_i\|^2 + \|V_i\|^2\big).$$

Applying Itô's formula gives

$$\begin{aligned} d\mathcal{E}_N = {}& \frac{2}{N}\sum_{i=1}^N \langle X_i, V_i\rangle\, dt + \frac{2}{N}\sum_{i=1}^N \langle V_i, -\nabla U(X_i) - \gamma V_i + F_i(X, V)\rangle\, dt \\ & + d\sigma^2\, dt + \frac{2\sigma}{N}\sum_{i=1}^N \langle V_i, dW_i\rangle. \end{aligned}$$

Use Young's inequality,

$$2\langle X_i, V_i\rangle \le \|X_i\|^2 + \|V_i\|^2, \qquad -2\langle V_i, \nabla U(X_i)\rangle \le \frac{1}{m_U}\|V_i\|^2 + m_U\|\nabla U(X_i)\|^2,$$

and the linear-growth bound

$$\|\nabla U(x)\|^2 \le 2L_U^2\|x\|^2 + 2\|\nabla U(0)\|^2.$$

For the viscous term, Definition 2.4 and $\omega_{ij} \le k^*/((N-1)k_*)$ imply

$$\begin{aligned}
\sum_{i=1}^N \langle V_i, F_i(X,V)\rangle &\le \nu_{\text{visc}} \sum_{i=1}^N |V_i| \left(\sum_{j\neq i} \omega_{ij}(X)|V_j| + |V_i|\right) \\
&\le \nu_{\text{visc}} \sum_{i=1}^N |V_i|^2 + \frac{\nu_{\text{visc}}}{2} \sum_{i=1}^N \sum_{j\neq i} \omega_{ij}(X)\big(|V_i|^2 + |V_j|^2\big) \\
&\le \nu_{\text{visc}} \left(\frac{3}{2} + \frac{k^*}{2k_*}\right) \sum_{i=1}^N |V_i|^2.
\end{aligned}$$

Consequently

$$\frac{d}{dt}\mathbb{E}\mathcal{E}_N(t) \le C_0 + C_1\,\mathbb{E}\mathcal{E}_N(t)$$

with constants depending only on $(m_U, L_U, \gamma, \sigma, \nu_{\text{visc}}, k_*, k^*)$ and not on N. Grönwall's lemma then gives $\sup_{t\le T}\mathbb{E}\mathcal{E}_N(t) < \infty$ on every finite interval, which excludes explosion. Hence the strong solution is global.

For the McKean–Vlasov equation, Lemma 3.4 gives the local Lipschitz dependence of the drift on (x, v, μ) on moment-bounded classes, while the force has at most linear growth in (x, v) under a finite second moment assumption on μ. A Picard iteration on the nonlinear SDE

$$d\bar{X}_t = \bar{V}_t\,dt, \qquad d\bar{V}_t = \big(-\nabla U(\bar{X}_t) - \gamma\bar{V}_t + F[f_t](\bar{X}_t, \bar{V}_t)\big)\,dt + \sigma\,dW_t$$

combined with the same Lyapunov estimate as above yields a unique global strong solution with law $f_t \in C([0,\infty); \mathcal{P}_2(\mathbb{R}^{2d}))$. □

Proposition 3.6 (Quadratic moment bounds). *Let $(X_i, V_i)_{i=1}^N$ solve (1)–(2) and let f_t solve (5). Then for every finite horizon $T > 0$ there exists $C_T < \infty$, depending only on the model parameters and the initial second moments, such that*

$$\sup_{t\in[0,T]} \frac{1}{N}\sum_{i=1}^N \mathbb{E}\big[\|X_i(t)\|^2 + \|V_i(t)\|^2\big] \le C_T$$

for every $N \ge 2$, and

$$\sup_{t\in[0,T]} \int_{\mathbb{R}^{2d}} (\|x\|^2 + \|v\|^2)\, f_t(dx, dv) \le C_T.$$

Proof. For the finite-particle system, the estimate from Proposition 3.5 already gives a closed differential inequality for the averaged quadratic energy, so the uniform bound follows by taking expectations and applying Grönwall's lemma.

For the mean-field equation, apply Itô's formula to $\|\bar{X}_t\|^2 + \|\bar{V}_t\|^2$ along the nonlinear SDE. Since

$$\|F[f_t](x,v)\| \le \frac{\nu_{\mathrm{visc}} k^*}{k_*} \left(\int \|q\| \, f_t(dy,dq) + \|v\| \right),$$

Young's inequality gives

$$\frac{d}{dt}\mathbb{E}\big[\|\bar{X}_t\|^2 + \|\bar{V}_t\|^2\big] \le C_0 + C_1 \mathbb{E}\big[\|\bar{X}_t\|^2 + \|\bar{V}_t\|^2\big].$$

Applying Grönwall's lemma yields the claimed bound. $\square$

4 Detailed Backbone and Viscous Derivations

4.1 Weighted generator calculus for the uncoupled kinetic backbone

For the detailed hypocoercive calculations it is convenient to work first with the uncoupled kinetic backbone. Set

$$T := \frac{\sigma^2}{2\gamma}$$

and define the Gibbs density

$$\rho_*(x,v) := Z_*^{-1} \exp\left(-\frac{U(x) + \frac{1}{2}\|v\|^2}{T} \right). \tag{7}$$

Under the smooth confining assumptions on U, this is the invariant density of the one-particle underdamped Langevin diffusion.

Definition 4.1 (Relative density and hypocoercive functionals)**.** Let f_t solve the uncoupled one-particle Fokker–Planck equation

$$\partial_t f_t + v \cdot \nabla_x f_t + \nabla_v \cdot \big((\nabla U(x) + \gamma v) f_t \big) = \frac{\sigma^2}{2} \Delta_v f_t,$$

and write

$$h_t(x,v) := \frac{f_t(x,v)}{\rho_*(x,v)}.$$

For smooth positive h define

$$\mathcal{A}[h] := \int h \log h \, d\rho_*, \tag{8}$$

$$\mathcal{B}[h] := T \int \frac{|\nabla_v h|^2}{h} \, d\rho_* = T \int h \, |\nabla_v \log h|^2 \, d\rho_*, \tag{9}$$

$$\mathcal{C}[h] := T \int \frac{\nabla_x h \cdot \nabla_v h}{h} \, d\rho_* = T \int h \, \langle \nabla_x \log h, \nabla_v \log h \rangle \, d\rho_*, \tag{10}$$

$$\mathcal{D}[h] := T \int \frac{|\nabla_x h|^2}{h} \, d\rho_* = T \int h \, |\nabla_x \log h|^2 \, d\rho_*. \tag{11}$$

Proposition 4.2 (Exact generator decomposition)**.** *Let L_0 denote the generator acting on h through*

$$\partial_t h = L_0 h = \rho_*^{-1} \partial_t (h \rho_*).$$

Then

$$L_0 = L_{\mathrm{sym}} + L_{\mathrm{anti}},$$

where

$$L_{\mathrm{sym}}h = \gamma\big(T\Delta_v h - v\cdot\nabla_v h\big), \tag{12}$$

$$L_{\mathrm{anti}}h = -v\cdot\nabla_x h + \nabla U(x)\cdot\nabla_v h. \tag{13}$$

Moreover:

(i) L_{sym} *is symmetric and non-positive in* $L^2(\rho_*)$*;*

(ii) L_{anti} *is anti-symmetric in* $L^2(\rho_*)$*.*

Proof. By definition,

$$L_0 h = \frac{1}{\rho_*}\left[-v\cdot\nabla_x(h\rho_*) + \nabla_v\cdot\big((\nabla U+\gamma v)h\rho_*\big) + \frac{\sigma^2}{2}\Delta_v(h\rho_*)\right].$$

We first expand the diffusion–friction part. Using

$$\nabla_v\rho_* = -\frac{v}{T}\rho_*, \qquad \frac{\sigma^2}{2} = \gamma T,$$

we obtain

$$\begin{aligned}\nabla_v\cdot\big((\nabla U+\gamma v)h\rho_*\big) + \gamma T\Delta_v(h\rho_*) &= \nabla U\cdot\nabla_v(h\rho_*) + \gamma\nabla_v\cdot(vh\rho_*) + \gamma T\Delta_v(h\rho_*)\\ &= \nabla U\cdot\nabla_v(h\rho_*) + \gamma\rho_*\, v\cdot\nabla_v h + \gamma d\, h\rho_* + \gamma v\cdot\nabla_v\rho_*\, h\\ &\quad + \gamma T\Delta_v(h\rho_*).\end{aligned}$$

The terms involving $vh\rho_*$ cancel after expanding $\Delta_v(h\rho_*)$ and using $\nabla_v\rho_* = -T^{-1}v\rho_*$. What remains is

$$\nabla U\cdot\nabla_v(h\rho_*) + \gamma T\nabla_v\cdot(\rho_*\nabla_v h).$$

Dividing by ρ_* gives

$$\nabla U\cdot\nabla_v h + \gamma\big(T\Delta_v h - v\cdot\nabla_v h\big).$$

The transport part expands as

$$-\frac{1}{\rho_*}v\cdot\nabla_x(h\rho_*) = -v\cdot\nabla_x h - v\cdot\nabla_x(\log\rho_*)\,h.$$

Since $\nabla_x\log\rho_* = -T^{-1}\nabla U$, the last term is $T^{-1}(v\cdot\nabla U)h$, which cancels exactly with the corresponding term created when expanding $\nabla U\cdot\nabla_v(h\rho_*)/\rho_*$. This proves (12)–(13).

For symmetry, let g, h be smooth compactly supported functions. Then

$$\int g\, L_{\mathrm{sym}}h\, d\rho_* = \gamma T\int g\,\rho_*^{-1}\nabla_v\cdot(\rho_*\nabla_v h)\, d\rho_* = -\gamma T\int \nabla_v g\cdot\nabla_v h\, d\rho_*,$$

which is symmetric in (g,h) and non-positive on the diagonal.

For anti-symmetry, write

$$b(x,v) := (v, -\nabla U(x)).$$

Then

$$L_{\mathrm{anti}}h = -b\cdot\nabla_{x,v}h.$$

It is enough to prove that $\nabla_{x,v} \cdot (b\rho_*) = 0$. Indeed,

$$\nabla_{x,v} \cdot (b\rho_*) = v \cdot \nabla_x \rho_* - \nabla U \cdot \nabla_v \rho_* = -\frac{1}{T}(v \cdot \nabla U)\rho_* + \frac{1}{T}(\nabla U \cdot v)\rho_* = 0.$$

Therefore

$$\int g L_{\mathrm{anti}} h \, d\rho_* = -\int g \, b \cdot \nabla h \, \rho_* = \int h \, b \cdot \nabla g \, \rho_* = -\int h L_{\mathrm{anti}} g \, d\rho_*. \qquad \square$$

Proposition 4.3 (Commutators)**.** *For smooth h,*

$$[\nabla_v, L_0]h = -\gamma \nabla_v h - \nabla_x h, \tag{14}$$

$$[\nabla_x, L_0]h = (\nabla^2 U)\nabla_v h. \tag{15}$$

Proof. The symmetric part depends only on v, so $[\nabla_x, L_{\mathrm{sym}}] = 0$ and

$$[\nabla_v, L_{\mathrm{sym}}]h = \gamma\big(T\nabla_v \Delta_v h - \nabla_v(v \cdot \nabla_v h) + v \cdot \nabla_v \nabla_v h\big) = -\gamma \nabla_v h.$$

For the anti-symmetric part,

$$[\nabla_v, -v \cdot \nabla_x]h = -\nabla_x h, \qquad [\nabla_v, \nabla U \cdot \nabla_v]h = 0,$$

which yields (14). Likewise,

$$[\nabla_x, -v \cdot \nabla_x]h = 0, \qquad [\nabla_x, \nabla U \cdot \nabla_v]h = (\nabla^2 U)\nabla_v h,$$

which is (15). $\square$

Proposition 4.4 (Exact differential identities)**.** *Let h_t solve $\partial_t h_t = L_0 h_t$. Set*

$$M := \sup_{x \in \mathbb{R}^d} \|\nabla^2 U(x)\|_{\mathrm{op}} \le L_U.$$

Then:

$$\frac{d}{dt}\mathcal{A}[h_t] = -\gamma \mathcal{B}[h_t], \tag{16}$$

$$\frac{d}{dt}\mathcal{B}[h_t] \le -2\gamma \mathcal{B}[h_t] - 2\mathcal{C}[h_t], \tag{17}$$

$$\frac{d}{dt}\mathcal{C}[h_t] \le -\mathcal{D}[h_t] - \gamma \mathcal{C}[h_t] + M\sqrt{\mathcal{B}[h_t]\mathcal{D}[h_t]}, \tag{18}$$

$$\frac{d}{dt}\mathcal{D}[h_t] \le 2M\sqrt{\mathcal{B}[h_t]\mathcal{D}[h_t]}. \tag{19}$$

Proof. Set $\ell_t := \log h_t$.

For (16), the anti-symmetric part drops out by Proposition 4.2(ii), so

$$\frac{d}{dt}\mathcal{A}[h_t] = \int (1 + \log h_t) L_{\mathrm{sym}} h_t \, d\rho_* = -\gamma T \int \frac{|\nabla_v h_t|^2}{h_t} \, d\rho_* = -\gamma \mathcal{B}[h_t].$$

We next differentiate

$$\mathcal{B}[h_t] = T \int h_t |\nabla_v \ell_t|^2 \, d\rho_*.$$

Since $\partial_t \ell_t = h_t^{-1} L_0 h_t$, the chain rule gives

$$\frac{d}{dt}\mathcal{B}[h_t] = \underbrace{T \int L_0 h_t \, |\nabla_v \ell_t|^2 \, d\rho_*}_{=:I_1} + \underbrace{2T \int h_t \, \nabla_v \ell_t \cdot \nabla_v (\partial_t \ell_t) \, d\rho_*}_{=:I_2} .$$

Because

$$\nabla_v (\partial_t \ell_t) = \nabla_v \left(\frac{L_0 h_t}{h_t} \right) = \frac{\nabla_v L_0 h_t}{h_t} - \frac{L_0 h_t}{h_t} \nabla_v \ell_t,$$

we obtain

$$I_2 = 2T \int \nabla_v \ell_t \cdot \nabla_v L_0 h_t \, d\rho_* - 2T \int L_0 h_t \, |\nabla_v \ell_t|^2 \, d\rho_*.$$

Therefore

$$\frac{d}{dt}\mathcal{B}[h_t] = 2T \int \nabla_v \ell_t \cdot \nabla_v L_0 h_t \, d\rho_* - T \int L_0 h_t \, |\nabla_v \ell_t|^2 \, d\rho_*.$$

Split $L_0 = L_{\mathrm{sym}} + L_{\mathrm{anti}}$ and treat both pieces separately.

For the symmetric piece, use

$$L_{\mathrm{sym}} = \gamma T \nabla_v^* \nabla_v,$$

where ∇_v^* is the $L^2(\rho_*)$ adjoint of ∇_v. A direct Bakry–Émery computation yields

$$-2\gamma T^2 \int h_t \|\nabla_v^2 \log h_t\|_{\mathrm{HS}}^2 \, d\rho_* \le 0,$$

and the first-order contribution is exactly $-2\gamma \mathcal{B}[h_t]$.

For the anti-symmetric piece, integrate by parts using Proposition 4.2(ii):

$$\begin{aligned} 2T \int \nabla_v \ell_t \cdot \nabla_v L_{\mathrm{anti}} h_t \, d\rho_* - T \int L_{\mathrm{anti}} h_t \, |\nabla_v \ell_t|^2 \, d\rho_* \\ = 2T \int h_t \, \nabla_v \ell_t \cdot [\nabla_v, L_{\mathrm{anti}}] \ell_t \, d\rho_*. \end{aligned}$$

By (14), $[\nabla_v, L_{\mathrm{anti}}] = -\nabla_x$, hence this term equals $-2\mathcal{C}[h_t]$. Combining the symmetric and anti-symmetric pieces gives

$$\frac{d}{dt}\mathcal{B}[h_t] = -2\gamma \mathcal{B}[h_t] - 2\mathcal{C}[h_t] - 2\gamma T^2 \int h_t \|\nabla_v^2 \log h_t\|_{\mathrm{HS}}^2 \, d\rho_*,$$

and (17) follows by discarding the final non-positive term.

We now turn to

$$\mathcal{C}[h_t] = T \int h_t \langle \nabla_x \ell_t, \nabla_v \ell_t \rangle \, d\rho_*.$$

Differentiating and grouping terms gives

$$\begin{aligned} \frac{d}{dt}\mathcal{C}[h_t] = T \int L_0 h_t \, \langle \nabla_x \ell_t, \nabla_v \ell_t \rangle \, d\rho_* \\ + T \int h_t \langle \nabla_x (\partial_t \ell_t), \nabla_v \ell_t \rangle \, d\rho_* \\ + T \int h_t \langle \nabla_x \ell_t, \nabla_v (\partial_t \ell_t) \rangle \, d\rho_*. \end{aligned}$$

Proceed exactly as for $\mathcal{B}$: rewrite $\partial_t \ell_t = h_t^{-1} L_0 h_t$, integrate by parts, and use the symmetry of L_{sym} and anti-symmetry of L_{anti}. All second-order terms from L_{sym} contribute a remainder $\mathcal{R}_C(t) \le 0$. The commutator terms are the only first-order contributions, so

$$\begin{aligned}\frac{d}{dt}\mathcal{C}[h_t] \le T \int h_t \langle \nabla_x \ell_t, [\nabla_v, L_0]\ell_t\rangle \, d\rho_* \\ + T \int h_t \langle \nabla_v \ell_t, [\nabla_x, L_0]\ell_t\rangle \, d\rho_*.\end{aligned}$$

Applying (14)–(15) gives

$$\frac{d}{dt}\mathcal{C}[h_t] \le -\mathcal{D}[h_t] - \gamma \mathcal{C}[h_t] + T \int h_t \langle \nabla_x \log h_t, (\nabla^2 U)\nabla_v \log h_t\rangle \, d\rho_*.$$

Using the operator bound on $\nabla^2 U$ and Cauchy–Schwarz,

$$\left| T \int h_t \langle \nabla_x \log h_t, (\nabla^2 U)\nabla_v \log h_t\rangle \, d\rho_* \right| \le M\sqrt{\mathcal{B}[h_t]\mathcal{D}[h_t]},$$

which gives (18).

Finally, starting from

$$\mathcal{D}[h_t] = T \int h_t |\nabla_x \ell_t|^2 \, d\rho_*,$$

the same differentiation yields

$$\frac{d}{dt}\mathcal{D}[h_t] = 2T \int h_t \langle \nabla_x \log h_t, [\nabla_x, L_0] \log h_t\rangle \, d\rho_* + \mathcal{R}_D(t),$$

with $\mathcal{R}_D(t) \le 0$ from the symmetric part, because L_{sym} acts only in velocity. Invoking (15) and Cauchy–Schwarz yields (19). □

Theorem 4.5 (Detailed backbone hypocoercive decay)**.** *Let $M = L_U$ and define*

$$\varepsilon := \min\left\{1, \frac{\gamma}{16(M+\gamma)^2}\right\}, \qquad a := \frac{\gamma\varepsilon}{4}, \qquad b := \varepsilon, \qquad c := \varepsilon^2.$$

For the modified entropy

$$\Phi[h] := \mathcal{A}[h] + a\mathcal{B}[h] + 2b\mathcal{C}[h] + c\mathcal{D}[h], \tag{20}$$

there exists $\lambda_{\text{back}} > 0$, depending only on (γ, M), such that

$$\frac{d}{dt}\Phi[h_t] \le -\lambda_{\text{back}}\big(\mathcal{B}[h_t] + \mathcal{D}[h_t]\big).$$

If the Gibbs measure ρ_* satisfies an LSI with constant $\rho_{\text{LSI}} > 0$, that is,

$$\rho_{\text{LSI}}\mathcal{A}[h] \le \mathcal{B}[h] + \mathcal{D}[h],$$

then

$$\Phi[h_t] \le e^{-\lambda_{\text{back}}\rho_{\text{LSI}} t}\Phi[h_0].$$

Proof. Insert (16)–(19) into the derivative of (20). This gives

$$\frac{d}{dt}\Phi[h_t] \le -\gamma\mathcal{B} + a(-2\gamma\mathcal{B} - 2\mathcal{C}) + 2b(-\mathcal{D} - \gamma\mathcal{C} + M\sqrt{\mathcal{B}\mathcal{D}}) \\ + c(2M\sqrt{\mathcal{B}\mathcal{D}}).$$

Using $-\mathcal{C} \le \sqrt{\mathcal{B}\mathcal{D}}$, we obtain

$$\frac{d}{dt}\Phi[h_t] \le -\gamma(1+2a)\mathcal{B} - 2b\mathcal{D} + \Xi\sqrt{\mathcal{B}\mathcal{D}},$$

where

$$\Xi := 2a + 2b\gamma + 2M(b+c).$$

Set $Y = (\sqrt{\mathcal{B}}, \sqrt{\mathcal{D}})^\top$. The right-hand side is $-Y^\top K Y$ for

$$K = \begin{pmatrix} \gamma(1+2a) & -\Xi/2 \\ -\Xi/2 & 2b \end{pmatrix}.$$

Since a, b, c are positive, $K_{11} \ge \gamma$ and $K_{22} = 2\varepsilon$. Moreover,

$$\Xi \le \frac{\gamma^2\varepsilon}{2} + 2\gamma\varepsilon + 2(M+\gamma)\varepsilon \le 4(M+\gamma)\varepsilon,$$

because $\varepsilon \le 1$. Hence

$$\det K \ge 2\gamma\varepsilon - 4(M+\gamma)^2\varepsilon^2 \ge \gamma\varepsilon > 0$$

by the choice of ε. Therefore K is positive definite and

$$Y^\top K Y \ge \lambda_{\min}(K)(\mathcal{B} + \mathcal{D})$$

where

$$\lambda_{\min}(K) = \frac{\gamma(1+2a) + 2b - \sqrt{\big(\gamma(1+2a) - 2b\big)^2 + \Xi^2}}{2}.$$

with

$$\lambda_{\text{back}} := \lambda_{\min}(K) > 0.$$

This proves the first claim. If ρ_* satisfies the stated LSI, then

$$\mathcal{B}[h_t] + \mathcal{D}[h_t] \ge \rho_{\text{LSI}}\mathcal{A}[h_t].$$

Since Φ is equivalent to $\mathcal{A} + \mathcal{B} + \mathcal{D}$ for fixed (a, b, c), the differential inequality closes and Grönwall's lemma gives the exponential decay of Φ. □

Proposition 4.6 (Gibbs logarithmic Sobolev inequality)**.** *Let*

$$W(x, v) := U(x) + \frac{1}{2}\|v\|^2, \qquad m_* := \min\{m_U, 1\}.$$

Then the Gibbs measure ρ_* from (7) *satisfies*

$$\operatorname{Ent}_{\rho_*}(h) \le \frac{1}{2m_*}\big(\mathcal{B}[h] + \mathcal{D}[h]\big) \tag{21}$$

for every smooth density $h \ge 0$ with $\int h\, d\rho_* = 1$.

Proof. The density of ρ_* has the form

$$\rho_*(x, v) = Z_*^{-1} e^{-W(x,v)/T}.$$

The Hessian of W is block diagonal:

$$\nabla^2 W(x, v) = \begin{pmatrix} \nabla^2 U(x) & 0 \\ 0 & I_d \end{pmatrix}.$$

Assumption 2.1(ii) therefore implies

$$\nabla^2 W(x, v) \succeq m_* I_{2d} \qquad \forall (x, v) \in \mathbb{R}^{2d}.$$

The Bakry–Émery criterion applied to the probability measure $Z_*^{-1} e^{-W/T} dx\, dv$ yields

$$\mathrm{Ent}_{\rho_*}(g^2) \leq \frac{2T}{m_*} \int |\nabla g|^2 \, d\rho_*$$

for every smooth g. Choose $g = \sqrt{h}$. Since

$$|\nabla \sqrt{h}|^2 = \frac{|\nabla_x h|^2 + |\nabla_v h|^2}{4h},$$

we obtain

$$\mathrm{Ent}_{\rho_*}(h) \leq \frac{T}{2m_*} \int \frac{|\nabla_x h|^2 + |\nabla_v h|^2}{h} \, d\rho_* = \frac{1}{2m_*} \big(\mathcal{B}[h] + \mathcal{D}[h] \big),$$

which is (21). □

Corollary 4.7 (One-particle backbone entropy contraction). *Let $P_t^{(1),0}$ be the semigroup of the uncoupled one-particle kinetic backbone and let $\pi_1 = \rho_*$. There exists a constant $\lambda_{\mathrm{hypo}} > 0$, depending only on $(m_U, L_U, \gamma, \sigma)$, such that*

$$\mathrm{KL}(\mu P_t^{(1),0} \,\|\, \pi_1) \leq e^{-2\lambda_{\mathrm{hypo}} t} \mathrm{KL}(\mu \,\|\, \pi_1)$$

for every probability measure $\mu \ll \pi_1$ and every $t \geq 0$.

Proof. Let $h_t = d(\mu P_t^{(1),0})/d\pi_1$. Proposition 4.6 yields

$$\mathcal{A}[h] \leq \frac{1}{2m_*} \big(\mathcal{B}[h] + \mathcal{D}[h] \big).$$

By Cauchy–Schwarz,

$$2|\mathcal{C}[h]| \leq \mathcal{B}[h] + \mathcal{D}[h].$$

Therefore the quadratic part of Φ satisfies

$$\Phi[h] \leq \left(\frac{1}{2m_*} + a + b + c \right) \big(\mathcal{B}[h] + \mathcal{D}[h] \big) =: C_\Phi \big(\mathcal{B}[h] + \mathcal{D}[h] \big).$$

Theorem 4.5 gives

$$\frac{d}{dt} \Phi[h_t] \leq -\lambda_{\mathrm{back}} \big(\mathcal{B}[h_t] + \mathcal{D}[h_t] \big) \leq -\frac{\lambda_{\mathrm{back}}}{C_\Phi} \Phi[h_t].$$

Hence

$$\Phi[h_t] \leq e^{-(\lambda_{\mathrm{back}}/C_\Phi) t} \Phi[h_0].$$

Since $\mathcal{A}[h_t] \le \Phi[h_t]$ and

$$\Phi[h_0] \le \left(1 + \frac{a+c+2b}{2m_*}\right)\big(\mathcal{B}[h_0] + \mathcal{D}[h_0]\big),$$

the same differential inequality applied from time s onward and integrated over $[s, \infty)$ gives

$$\mathcal{A}[h_s] = \gamma \int_s^\infty \mathcal{B}[h_t]\, dt \le \gamma \int_s^\infty \big(\mathcal{B}[h_t] + \mathcal{D}[h_t]\big)\, dt \le \frac{\gamma C_\Phi}{\lambda_{\text{back}}}\Phi[h_s].$$

Using $\mathcal{A}[h_s] \le \Phi[h_s]$ once more and absorbing constants yields an exponential bound

$$\mathcal{A}[h_t] \le e^{-2\lambda_{\text{hypo}} t}\mathcal{A}[h_0]$$

for some $\lambda_{\text{hypo}} > 0$ depending only on $(m_U, L_U, \gamma, \sigma)$. This is the stated entropy contraction. □

4.2 Matrix analysis for the viscous velocity block

Definition 4.8 (Viscous matrices)**.** For $X = (x_1, \dots, x_N)$ define

$$K_{ij}(X) := \begin{cases} K(x_i, x_j), & i \ne j, \\ 0, & i = j, \end{cases} \qquad D(X) := \operatorname{diag}(d_1(X), \dots, d_N(X)),$$

and

$$W(X) := D(X)^{-1}K(X), \qquad L(X) := I - W(X).$$

The velocity dynamics may then be written as

$$dV = \big(-\nabla U(X) - A(X)V\big)\, dt + \sigma\, dW, \qquad A(X) := \gamma I_{Nd} + \nu_{\text{visc}}\, L(X) \otimes I_d.$$

Lemma 4.9 (Weighted symmetry and spectral bounds)**.** *For every configuration X:*

(i) $D(X)W(X) = K(X)$ *is symmetric;*

(ii) *the matrix*

$$\widetilde{L}(X) := D(X)^{1/2}L(X)D(X)^{-1/2} = D(X)^{-1/2}(D(X) - K(X))D(X)^{-1/2}$$

is symmetric positive semidefinite;

(iii) *every eigenvalue of $\widetilde{L}(X)$ lies in $[0, 2]$;*

(iv) $A(X)$ *is similar to the symmetric positive definite matrix*

$$\widetilde{A}(X) := \gamma I_{Nd} + \nu_{\text{visc}}\widetilde{L}(X) \otimes I_d,$$

hence every eigenvalue of $A(X)$ lies in $[\gamma, \gamma + 2\nu_{\text{visc}}]$.

Proof. The identity $D(X)W(X) = K(X)$ is immediate from the definitions, and $K(X)$ is symmetric because the kernel is symmetric. Therefore

$$\widetilde{L}(X) = D(X)^{-1/2}(D(X) - K(X))D(X)^{-1/2}$$

is symmetric. For any $z \in \mathbb{R}^N$,

$$z^\top (D-K)z = \frac{1}{2}\sum_{i\neq j} K_{ij}(z_i - z_j)^2 \geq 0,$$

which proves positive semidefiniteness.

To bound the top of the spectrum, let $y \in \mathbb{R}^N$ and set $z = D(X)^{-1/2}y$. Then

$$y^\top \widetilde{L}(X)y = z^\top (D-K)z = \frac{1}{2}\sum_{i\neq j} K_{ij}(z_i - z_j)^2.$$

Also,

$$y^\top y = \sum_i d_i z_i^2.$$

Using $(z_i - z_j)^2 \leq 2z_i^2 + 2z_j^2$ and the symmetry of K,

$$y^\top \widetilde{L}(X)y \leq \sum_{i\neq j} K_{ij}(z_i^2 + z_j^2) = 2\sum_i d_i z_i^2 = 2\, y^\top y.$$

Since $\widetilde{L}(X)$ is symmetric positive semidefinite, its Rayleigh quotient belongs to $[0,2]$, and hence every eigenvalue lies in $[0,2]$.

Finally,

$$A(X) = \big(D(X)^{-1/2} \otimes I_d\big)\widetilde{A}(X)\big(D(X)^{1/2} \otimes I_d\big),$$

so $A(X)$ and $\widetilde{A}(X)$ are similar and hence share the same spectrum. Since $\widetilde{L}(X)$ is symmetric positive semidefinite with spectrum in $[0,2]$, the spectrum of $\widetilde{A}(X)$ is contained in $[\gamma, \gamma + 2\nu_{\text{visc}}]$. $\square$

Proposition 4.10 (Frozen-position Gaussian law and covariance bound)**.** *Fix a configuration X and consider the frozen-position velocity process*

$$dV_t = \big(-b(X) - A(X)V_t\big)\,dt + \sigma\, dW_t, \qquad b(X) := (\nabla U(x_1), \dots, \nabla U(x_N)).$$

Then:

(i) *the stationary law is Gaussian*

$$\mathcal{N}\big(m_X, \Sigma_X\big), \qquad m_X = -A(X)^{-1}b(X);$$

(ii) *the covariance solves the Lyapunov equation*

$$A(X)\Sigma_X + \Sigma_X A(X)^\top = \sigma^2 I_{Nd}; \tag{22}$$

(iii) *its largest eigenvalue satisfies the N-uniform bound*

$$\lambda_{\max}(\Sigma_X) \leq \frac{k^*}{k_*}\frac{\sigma^2}{2\gamma}.$$

Proof. For fixed X, the velocity equation is a linear Ornstein–Uhlenbeck process with constant drift matrix $A(X)$ and constant forcing $-b(X)$. By Lemma 4.9(iv), the spectrum of $A(X)$ lies in $[\gamma, \infty)$, so the linear system is exponentially stable. The unique stationary law is therefore Gaussian with mean

$$m_X = -A(X)^{-1}b(X),$$

and the covariance Σ_X is the unique positive semidefinite solution of (22).

To bound the covariance, write $H(X) := D(X)^{1/2} \otimes I_d$ and

$$A(X) = H(X)^{-1}\widetilde{A}(X)H(X), \qquad \widetilde{A}(X) = \gamma I_{Nd} + \nu_{\mathrm{visc}}\widetilde{L}(X) \otimes I_d.$$

Since $\widetilde{A}(X)$ is symmetric positive definite, the covariance admits the standard OU representation

$$\Sigma_X = \sigma^2 \int_0^\infty e^{-A(X)t} e^{-A(X)^\top t}\, dt.$$

Substituting the similarity transform gives

$$\Sigma_X = \sigma^2 H^{-1} \left[\int_0^\infty e^{-\widetilde{A}t} H^2 e^{-\widetilde{A}t}\, dt\right] H^{-1}.$$

For any $u \in \mathbb{R}^{Nd}$ with $\|u\| = 1$, set $w = H^{-1}u$. Then

$$\begin{aligned} u^\top \Sigma_X u &= \sigma^2 \int_0^\infty \langle He^{-\widetilde{A}t}w,\, He^{-\widetilde{A}t}w\rangle\, dt \\ &\le \sigma^2 \lambda_{\max}(H^2) \int_0^\infty \|e^{-\widetilde{A}t}w\|^2\, dt \\ &\le \sigma^2 \lambda_{\max}(H^2) \int_0^\infty e^{-2\gamma t}\, dt\, \|w\|^2 \\ &\le \frac{\sigma^2}{2\gamma} \frac{\lambda_{\max}(H^2)}{\lambda_{\min}(H^2)}. \end{aligned}$$

Because

$$(N-1)k_* \le d_i(X) \le (N-1)k^* \qquad \forall i,$$

we have

$$\frac{\lambda_{\max}(H^2)}{\lambda_{\min}(H^2)} = \frac{\max_i d_i(X)}{\min_i d_i(X)} \le \frac{k^*}{k_*}.$$

Taking the supremum over unit vectors u proves the stated bound on $\lambda_{\max}(\Sigma_X)$. $\square$

Corollary 4.11 (Conditional Gaussian Poincaré inequality)**.** *For every frozen configuration X and every smooth $g : \mathbb{R}^{Nd} \to \mathbb{R}$,*

$$\mathrm{Var}_{\mathcal{N}(m_X,\Sigma_X)}(g) \le \frac{k^*}{k_*}\frac{\sigma^2}{2\gamma} \int |\nabla_v g|^2\, d\mathcal{N}(m_X, \Sigma_X).$$

Proof. For a centered Gaussian law $\mathcal{N}(m, \Sigma)$, the sharp Poincaré constant is $\lambda_{\max}(\Sigma)$. Since translating by the mean does not change the variance or the gradient term, the same constant applies to $\mathcal{N}(m_X, \Sigma_X)$. Proposition 4.10 bounds $\lambda_{\max}(\Sigma_X)$ by $\frac{k^*}{k_*}\frac{\sigma^2}{2\gamma}$, so the stated inequality follows. $\square$

Lemma 4.12 (Derivative bounds for the normalized weights)**.** *Let*

$$L_K := \sup_{x,y\in\mathbb{R}^d} \|\nabla_x K(x,y)\| < \infty.$$

Then for every $N \geq 2$ and every $i \neq j$,

$$\|\nabla_{x_i}\omega_{ij}(X)\| \leq \frac{L_K}{(N-1)k_*} + \frac{k^* L_K}{(N-1)k_*^2},$$

and

$$\|\nabla_{x_j}\omega_{ij}(X)\| \leq \frac{L_K}{(N-1)k_*} + \frac{k^* L_K}{(N-1)^2 k_*^2},$$

and for $m \notin \{i,j\}$,

$$\|\nabla_{x_m}\omega_{ij}(X)\| \leq \frac{k^* L_K}{(N-1)^2 k_*^2}.$$

In particular,

$$\sup_{N\geq 2}\sup_{X}\max_{i=1,\ldots,N}\max_{m=1,\ldots,N}\sum_{j\neq i}\|\nabla_{x_m}\omega_{ij}(X)\| < \infty.$$

Proof. Differentiate

$$\omega_{ij}(X) = \frac{K(x_i,x_j)}{d_i(X)}, \qquad d_i(X) = \sum_{k\neq i} K(x_i,x_k).$$

For the x_i derivative,

$$\nabla_{x_i}\omega_{ij} = \frac{\nabla_{x_i}K(x_i,x_j)}{d_i} - \frac{K(x_i,x_j)}{d_i^2}\sum_{k\neq i}\nabla_{x_i}K(x_i,x_k).$$

Taking norms and using $d_i \geq (N-1)k_*$, $K(x_i,x_j) \leq k^*$, and $\|\nabla_{x_i}K\| \leq L_K$ yields

$$\|\nabla_{x_i}\omega_{ij}\| \leq \frac{L_K}{(N-1)k_*} + \frac{k^*(N-1)L_K}{(N-1)^2 k_*^2}.$$

This is the first estimate. For $m \notin \{i,j\}$ only the denominator depends on x_m, so

$$\nabla_{x_m}\omega_{ij} = -\frac{K(x_i,x_j)}{d_i^2}\nabla_{x_m}K(x_i,x_m),$$

which gives the third estimate. For the x_j derivative,

$$\nabla_{x_j}\omega_{ij} = \frac{\nabla_{x_j}K(x_i,x_j)}{d_i} - \frac{K(x_i,x_j)}{d_i^2}\nabla_{x_j}K(x_i,x_j),$$

and the same lower bound on d_i gives the second estimate. The uniform summability follows by summing the displayed bounds over j for fixed (i,m). □

Lemma 4.13 (Derivative bounds for the viscous force)**.** *There exists a constant $C_{\partial F} < \infty$, depending only on $(\nu_{\text{visc}}, k_*, k^*, L_K)$, such that for every configuration $(X,V) \in (\mathbb{R}^{2d})^N$:*

$$\max_{1\leq i\leq N}\|\nabla_V F_i(X,V)\|_{\text{op}} \leq 2\nu_{\text{visc}}, \tag{23}$$

$$\sum_{m=1}^{N}\|\nabla_{x_m}F_i(X,V)\|^2 \leq C_{\partial F}\left(\|V_i\|^2 + \frac{1}{N}\sum_{j=1}^{N}\|V_j\|^2\right). \tag{24}$$

Proof. The velocity derivatives are explicit:

$$\nabla_{v_i} F_i = -\nu_{\mathrm{visc}} I_d, \qquad \nabla_{v_m} F_i = \nu_{\mathrm{visc}} \omega_{im}(X) I_d \quad (m \neq i).$$

Hence

$$\|\nabla_V F_i\|_{\mathrm{op}} \le \nu_{\mathrm{visc}} + \nu_{\mathrm{visc}} \sum_{m \neq i} \omega_{im}(X) = 2\nu_{\mathrm{visc}},$$

which proves (23).

For the x derivatives,

$$\nabla_{x_m} F_i(X, V) = \nu_{\mathrm{visc}} \sum_{j \neq i} \nabla_{x_m} \omega_{ij}(X)\,(V_j - V_i).$$

Define

$$A_{im}(X) := \sum_{j \neq i} \|\nabla_{x_m} \omega_{ij}(X)\|.$$

By Lemma 4.12,

$$\sup_{N \ge 2} \sup_X \max_{i,m} A_{im}(X) \le C_\omega$$

for some finite constant C_ω. Using Cauchy–Schwarz in the discrete sum,

$$\begin{aligned}
\|\nabla_{x_m} F_i(X, V)\|^2 &\le \nu_{\mathrm{visc}}^2 \left(\sum_{j \neq i} \|\nabla_{x_m} \omega_{ij}\| \right) \left(\sum_{j \neq i} \|\nabla_{x_m} \omega_{ij}\|\, \|V_j - V_i\|^2 \right) \\
&\le \nu_{\mathrm{visc}}^2 C_\omega \left(\sum_{j \neq i} \|\nabla_{x_m} \omega_{ij}\|\, (2\|V_j\|^2 + 2\|V_i\|^2) \right).
\end{aligned}$$

Summing over m, and using the explicit derivative bounds from Lemma 4.12, we have for every $j \neq i$

$$\sum_{m=1}^{N} \|\nabla_{x_m} \omega_{ij}(X)\| \le \frac{C_\omega}{N},$$

because only the terms $m \in \{i, j\}$ are of order $(N-1)^{-1}$ while the remaining $(N-2)$ terms are of order $(N-1)^{-2}$. Summing over $j \neq i$, this yields

$$\sum_{m=1}^{N} \sum_{j \neq i} \|\nabla_{x_m} \omega_{ij}(X)\| \le C_\omega.$$

Then, by the same bound,

$$\sum_{m=1}^{N} \sum_{j \neq i} \|\nabla_{x_m} \omega_{ij}(X)\|\, \|V_j\|^2 \le C_\omega \frac{1}{N} \sum_{j=1}^{N} \|V_j\|^2.$$

Collecting terms yields (24). $\square$

5 Finite-Particle Hypocoercivity and N-Uniform LSI

5.1 Backbone gap and viscous perturbation

For the kinetic backbone ($\nu_{\text{visc}} = 0$), the invariant law is $\pi_1^{\otimes N}$ and the gap from Corollary 4.7 tensorizes immediately.

Proposition 5.1 (Tensorized backbone entropy gap). *Let $P_t^{N,0}$ denote the semigroup of the uncoupled N-particle kinetic backbone obtained by setting $\nu_{\text{visc}} = 0$ in Definition 2.5. Then*

$$\mathrm{KL}(\mu P_t^{N,0} \,\|\, \pi_1^{\otimes N}) \leq e^{-2\lambda_{\text{hypo}} t} \mathrm{KL}(\mu \,\|\, \pi_1^{\otimes N})$$

for every $\mu \ll \pi_1^{\otimes N}$ and every $N \geq 1$.

Proof. The uncoupled generator is the sum of the N one-particle generators, so the semigroup factorizes as

$$P_t^{N,0} = (P_t^{(1),0})^{\otimes N}.$$

Let $\mu \ll \pi_1^{\otimes N}$ and write

$$h_0 = \frac{d\mu}{d\pi_1^{\otimes N}}.$$

Let $h_t := \frac{d(\mu P_t^{N,0})}{d\pi_1^{\otimes N}}$. The relative entropy with respect to a product reference measure satisfies the usual conditional chain rule:

$$\mathrm{KL}(\mu P_t^{N,0} \,\|\, \pi_1^{\otimes N}) = \sum_{i=1}^{N} \mathbb{E}_{\mu P_t^{N,0}}\Big[\mathrm{KL}\big((\mu P_t^{N,0})_{i\,|\,1:i-1}(\cdot \mid Z_1, \ldots, Z_{i-1}) \,\|\, \pi_1\big)\Big].$$

Because $P_t^{N,0}$ acts independently on each coordinate and preserves π_1, the i-th conditional law evolves by the one-particle semigroup $P_t^{(1),0}$ while the earlier coordinates are frozen. Corollary 4.7 therefore gives, for every conditioning value,

$$\mathrm{KL}\big((\mu P_t^{N,0})_{i\,|\,1:i-1}(\cdot \mid Z_1, \ldots, Z_{i-1}) \,\|\, \pi_1\big) \leq e^{-2\lambda_{\text{hypo}} t} \mathrm{KL}\big(\mu_{i\,|\,1:i-1}(\cdot \mid Z_1, \ldots, Z_{i-1}) \,\|\, \pi_1\big).$$

Taking expectations and summing over i gives

$$\mathrm{KL}(\mu P_t^{N,0} \,\|\, \pi_1^{\otimes N}) \leq e^{-2\lambda_{\text{hypo}} t} \mathrm{KL}(\mu \,\|\, \pi_1^{\otimes N})$$

for every $\mu \ll \pi_1^{\otimes N}$ and every $N \geq 1$. $\square$

Remark 5.2 (Hypocoercive carré du champ). For the backbone dynamics one may choose a Villani-type quadratic form

$$\Gamma_{\text{hypo}}^N(g, g) := a\|\nabla_v g\|^2 + 2b\, \nabla_x g \cdot \nabla_v g + c\|\nabla_x g\|^2, \qquad ac > b^2,$$

with coefficients depending only on (U, γ, σ) and not on N, so that Γ_{hypo}^N is equivalent to the full H^1 norm under the backbone invariant law. The entropy gap of Proposition 5.1 is equivalent to an estimate of the form

$$\mathrm{Ent}_{\pi_1^{\otimes N}}(g^2) \leq C_{\text{hypo}} \int \Gamma_{\text{hypo}}^N(g, g)\, d\pi_1^{\otimes N}.$$

We now explain why the viscous alignment does not destroy this structure; the underlying modified-entropy mechanism is the one introduced in [7].

Definition 5.3 (Finite-particle entropy and Fisher functionals)**.** Let Π_N be a smooth probability measure on $(\mathbb{R}^{2d})^N$, and let $h : (\mathbb{R}^{2d})^N \to (0, \infty)$ satisfy $\int h \, d\Pi_N = 1$. Writing $X = (x_1, \dots, x_N)$ and $V = (v_1, \dots, v_N)$, define

$$\begin{aligned}
\mathcal{A}_N[h] &:= \int h \log h \, d\Pi_N, \\
\mathcal{B}_N[h] &:= T \int h \, |\nabla_V \log h|^2 \, d\Pi_N, \\
\mathcal{C}_N[h] &:= T \int h \, \langle \nabla_X \log h, \nabla_V \log h \rangle \, d\Pi_N, \\
\mathcal{D}_N[h] &:= T \int h \, |\nabla_X \log h|^2 \, d\Pi_N.
\end{aligned}$$

Here ∇_X and ∇_V denote the full gradients in the Nd position and velocity variables.

Theorem 5.4 (N-uniform hypocoercive LSI for the kinetic core)**.** *Assume that the full interacting diffusion* (1)–(2) *admits a unique invariant law π_N. Then there exist constants $\lambda_* > 0$ and $C_{\mathrm{LSI}} < \infty$, depending only on the backbone parameters and the kernel bounds, but* independent of N, *such that for every $N \geq 2$:*

$$\mathrm{Ent}_{\pi_N}(g^2) \leq C_{\mathrm{LSI}} \int \Gamma^N_{\mathrm{hypo}}(g, g) \; \mathrm{d}\pi_N, \tag{25}$$

$$\mathrm{KL}(\mu P^N_t \,\|\, \pi_N) \leq e^{-2\lambda_* t} \mathrm{KL}(\mu \,\|\, \pi_N) \qquad \forall \mu \ll \pi_N. \tag{26}$$

The constants λ_* and C_{LSI} depend only on $\gamma, \sigma, L_U, k_*, k^*, \ell_{\mathrm{visc}}$, and ν_{visc}, and are independent of N.

Proof. We isolate the two ingredients already established in detail.

Step 1: backbone hypocoercivity. For $\nu_{\mathrm{visc}} = 0$, Proposition 5.1 yields the tensorized entropy gap. The more explicit weighted computation is recorded in Theorem 4.5: the transport commutator is controlled by the Hessian bound $M = L_U$, and the modified entropy functional

$$\Phi_N[h] = \mathcal{A}_N[h] + a\mathcal{B}_N[h] + 2b\mathcal{C}_N[h] + c\mathcal{D}_N[h]$$

decays with a rate depending only on (γ, L_U, σ) and not on N.

Step 2: structure of the viscous perturbation. For the interacting dynamics, the velocity drift matrix is $A(X) = \gamma I + \nu_{\mathrm{visc}} L(X) \otimes I_d$. By Lemma 4.9, $A(X)$ is similar to a symmetric positive definite matrix with spectrum contained in $[\gamma, \gamma + 2\nu_{\mathrm{visc}}]$, uniformly in N and in the configuration X. Therefore the velocity damping remains uniformly accretive. In addition, Proposition 4.10 and Corollary 4.11 give an N-uniform Gaussian control on the frozen-position velocity block. This is precisely the quantitative input needed for the velocity-direction coercivity in the hypocoercive closure.

Step 3: commutator control for the X-dependence of the weights. The only new commutators relative to the backbone come from the derivatives of the normalized weights. Lemma 4.12 shows that the derivatives of $\omega_{ij}(X)$ are bounded by constants depending only on (k_*, k^*, L_K), uniformly in N. By Lemma 4.13, there is a constant $C_{\mathrm{visc}} < \infty$, depending only on the same kernel data and ν_{visc}, such that each commutator identity in Proposition 4.4 picks up an additional remainder term bounded by

$$C_{\mathrm{visc}} \sqrt{\mathcal{B}_N[h] \mathcal{D}_N[h]}$$

for every N.

Step 4: closure of the interacting hypocoercive estimate. The detailed backbone calculation from Theorem 4.5 therefore survives with the replacements

$$M \longmapsto M + C_{\text{visc}}, \qquad \gamma \longmapsto \gamma,$$

because the viscous block strengthens the velocity dissipation and modifies only the mixed commutator constants. Choosing the coefficients a, b, c exactly as in Theorem 4.5, but with $M + C_{\text{visc}}$ in place of M, produces a positive definite dissipation matrix K_{int} whose smallest eigenvalue is bounded below by a constant $\lambda_* > 0$ independent of N.

Equivalently, the interacting modified entropy obeys

$$\frac{d}{dt}\Phi_N[h_t] \leq -\lambda_*\big(\mathcal{B}_N[h_t] + \mathcal{D}_N[h_t]\big).$$

To pass from this differential inequality to the logarithmic Sobolev estimate, let $h_t = d(\mu P_t^N)/d\pi_N$. The entropy production identity is

$$\frac{d}{dt}\text{Ent}_{\pi_N}(h_t) = -\gamma\mathcal{B}_N[h_t].$$

Since the quadratic form in $(\mathcal{B}_N, \mathcal{C}_N, \mathcal{D}_N)$ is positive definite, there exists $\alpha_0 > 0$, depending only on (a, b, c), such that

$$\Phi_N[h] \geq \text{Ent}_{\pi_N}(h) + \alpha_0\big(\mathcal{B}_N[h] + \mathcal{D}_N[h]\big).$$

Integrating the previous two displays over $[0, \infty)$ gives

$$\text{Ent}_{\pi_N}(h_0) = \gamma\int_0^\infty \mathcal{B}_N[h_t]\, dt \leq \frac{\gamma}{\alpha_0\lambda_*}\Phi_N[h_0].$$

It remains to control $\Phi_N[h_0]$ by the hypocoercive carré du champ. Write $h_0 = g^2/\int g^2\, d\pi_N$. Then

$$\mathcal{B}_N[h_0] = 4T\int |\nabla_v g|^2\, d\pi_N, \qquad \mathcal{D}_N[h_0] = 4T\int |\nabla_x g|^2\, d\pi_N,$$

and

$$|\mathcal{C}_N[h_0]| \leq 4T\left(\int |\nabla_x g|^2\, d\pi_N\right)^{1/2}\left(\int |\nabla_v g|^2\, d\pi_N\right)^{1/2}.$$

The conditional Gaussian Poincaré inequality from Corollary 4.11 gives the microscopic coercivity in velocity, while the confining potential and the transport commutator provide the macroscopic coercivity in position. Combining these estimates with the previous display yields

$$\Phi_N[h_0] \leq C_0\int \Gamma^N_{\text{hypo}}(g, g)\, d\pi_N$$

for an N-independent constant C_0. Therefore

$$\text{Ent}_{\pi_N}(g^2) \leq \frac{\gamma C_0}{\alpha_0\lambda_*}\int \Gamma^N_{\text{hypo}}(g, g)\, d\pi_N.$$

This is (25) with $C_{\text{LSI}} = \gamma C_0/(\alpha_0\lambda_*)$. The entropy decay estimate (26) follows by applying (25) to $g = \sqrt{d(\mu P_t^N)/d\pi_N}$ and using Grönwall's lemma. $\square$

Remark 5.5. Under the smooth coercive hypotheses stated here, the existence and uniqueness of the invariant law π_N can be obtained by Lyapunov, minorization, and Harris arguments for underdamped Langevin diffusions [5, 4]. The new point in the present model is that the normalized viscous perturbation is dissipative and therefore does not degrade the backbone hypocoercive gap.

Corollary 5.6 (Exponential convergence in total variation)**.** *Under the assumptions of Theorem 5.4,*

$$\|\mu P_t^N - \pi_N\|_{\mathrm{TV}} \le \sqrt{2}\, e^{-\lambda_* t}\, \mathrm{KL}(\mu \,\|\, \pi_N)^{1/2} \qquad \forall \mu \ll \pi_N.$$

Proof. Pinsker's inequality says that for any probability measures α, β,

$$\|\alpha - \beta\|_{\mathrm{TV}} \le \sqrt{2\, \mathrm{KL}(\alpha \| \beta)}.$$

Apply this with $\alpha = \mu P_t^N$ and $\beta = \pi_N$, then use (26):

$$\|\mu P_t^N - \pi_N\|_{\mathrm{TV}} \le \sqrt{2\, \mathrm{KL}(\mu P_t^N \| \pi_N)} \le \sqrt{2}\, e^{-\lambda_* t}\, \mathrm{KL}(\mu \| \pi_N)^{1/2}.$$

□

5.2 The squashed observation chain

Proposition 5.7 (The squashing step cannot enlarge probabilistic distances)**.** *Let Q_τ^N be the time-τ observation operator obtained by evolving the continuous diffusion for time τ and then applying the deterministic map $S(X, V) = (X, \psi(V))$ componentwise. Then for every pair of probability measures μ, ν on $(\mathbb{R}^{2d})^N$,*

$$\|S_\# \mu - S_\# \nu\|_{\mathrm{TV}} \le \|\mu - \nu\|_{\mathrm{TV}}.$$

If W_p denotes any Wasserstein distance with $1 \le p < \infty$, then

$$W_p(S_\# \mu, S_\# \nu) \le W_p(\mu, \nu).$$

Proof. For total variation, if A is measurable in the target space then

$$(S_\# \mu)(A) - (S_\# \nu)(A) = \mu(S^{-1} A) - \nu(S^{-1} A),$$

so taking the supremum over measurable sets immediately yields $\|S_\# \mu - S_\# \nu\|_{\mathrm{TV}} \le \|\mu - \nu\|_{\mathrm{TV}}$.

For Wasserstein distances, let π be any coupling of μ and ν. Then $(S, S)_\# \pi$ is a coupling of $S_\# \mu$ and $S_\# \nu$, and Lemma 3.1 gives

$$\|(X, \psi(V)) - (X', \psi(V'))\| \le \|(X, V) - (X', V')\|.$$

Integrating this inequality against π and taking the infimum over couplings proves the Wasserstein contraction. □

Remark 5.8. Proposition 5.7 is the precise reason one can carry out the convergence proofs at the level of the unsquashed continuous SDE and only afterwards impose the algorithmic speed cap. The cap improves boundedness of the observed chain and never worsens the entropy, total-variation, or Wasserstein estimates.

6 Entropy and Total-Variation Convergence

The finite-particle entropy estimate controls the continuous interacting kinetic core. This section records the two consequences needed later: convergence of the observed chain and the existence of an N-independent relaxation rate.

Corollary 6.1 (Observed kinetic chain)**.** *Let Q_τ^N be the squashed observation chain from Proposition 5.7. If the continuous kinetic core satisfies* (26)*, then for every initial law $\mu \ll \pi_N$,*

$$\|\mu Q_\tau^{N,m} - S_\# \pi_N\|_{\mathrm{TV}} \le \sqrt{2}\, e^{-\lambda_* m\tau}\, \mathrm{KL}(\mu \,\|\, \pi_N)^{1/2} \qquad \forall m \in \mathbb{N}.$$

At all observation times the velocities also obey the deterministic bound $\|v_i\| \le V_{\mathrm{alg}}$.

Proof. Let $P_{m\tau}^N$ denote the continuous-time evolution over $m\tau$. The observation chain is obtained by composing that evolution with the deterministic map S, and both the Markov kernel $P_{m\tau}^N$ and the pushforward $S_\#$ are contractions in total variation. Therefore

$$\|\mu Q_\tau^{N,m} - S_\# \pi_N\|_{\mathrm{TV}} \le \|\mu P_{m\tau}^N - \pi_N\|_{\mathrm{TV}}.$$

Now apply Corollary 5.6 at time $m\tau$ to get the displayed bound. The velocity estimate follows directly from $\|\psi(v)\| \le V_{\mathrm{alg}}$, so every observed velocity is capped at the algorithmic bound. □

7 Mean-Field Equation and Propagation of Chaos

7.1 The McKean–Vlasov limit

We now compare the finite-particle law to the nonlinear mean-field equation (5). The key fact is that the interaction field from Definition 2.6 is globally Lipschitz in both the state variables and the underlying measure, thanks to the strictly positive kernel floor.

Lemma 7.1 (Empirical-force defect)**.** *Let*

$$\mu_t^N := \frac{1}{N} \sum_{j=1}^N \delta_{(X_j(t), V_j(t))}$$

be the empirical measure of the particle system and, for each i, let

$$F_i^N(t) := \nu_{\mathrm{visc}} \sum_{j \ne i} \omega_{ij}(X(t))(V_j(t) - V_i(t))$$

denote the exact finite-particle alignment force. Then for every finite horizon $T > 0$ there exists $C_T < \infty$, independent of N, such that

$$\sup_{t\in[0,T]} \frac{1}{N} \sum_{i=1}^N \mathbb{E}\Big[\big\|F_i^N(t) - F[\mu_t^N](X_i(t), V_i(t))\big\|^2\Big] \le \frac{C_T}{N}.$$

Proof. Fix i and abbreviate $X_i = X_i(t)$, $V_i = V_i(t)$. By definition,

$$F[\mu_t^N](X_i, V_i) = \frac{\nu_{\mathrm{visc}}}{m_{\mu_t^N}(X_i)} \frac{1}{N} \sum_{j=1}^N K(X_i, X_j)(V_j - V_i),$$

where

$$m_{\mu_t^N}(X_i) = \frac{1}{N}\sum_{j=1}^{N} K(X_i, X_j).$$

Since $K(X_i, X_i) = 1 + \kappa_0$, the finite-particle force and the empirical-field force differ only because of the self-interaction term and the normalization:

$$F_i^N - F[\mu_t^N](X_i, V_i) = \nu_{\mathrm{visc}}\left[\sum_{j\neq i}\frac{K(X_i, X_j)}{\sum_{k\neq i} K(X_i, X_k)}(V_j - V_i) - \frac{\sum_{j\neq i} K(X_i, X_j)(V_j - V_i)}{\sum_{k=1}^{N} K(X_i, X_k)}\right].$$

Write

$$S_i := \sum_{j\neq i} K(X_i, X_j)(V_j - V_i), \qquad d_i := \sum_{k\neq i} K(X_i, X_k).$$

Then

$$F_i^N - F[\mu_t^N](X_i, V_i) = \nu_{\mathrm{visc}} S_i\left(\frac{1}{d_i} - \frac{1}{d_i + K(X_i, X_i)}\right) = \nu_{\mathrm{visc}} S_i \frac{K(X_i, X_i)}{d_i(d_i + K(X_i, X_i))}.$$

By Assumption 2.3,

$$d_i \geq (N-1)k_*, \qquad K(X_i, X_i) \leq k^*,$$

and therefore

$$\left|\frac{K(X_i, X_i)}{d_i(d_i + K(X_i, X_i))}\right| \leq \frac{k^*}{(N-1)^2 k_*^2}.$$

On the other hand,

$$\|S_i\| \leq \sum_{j\neq i} K(X_i, X_j)\|V_j - V_i\| \leq 2k^* \sum_{j\neq i}(\|V_j\| + \|V_i\|).$$

Using the uniform second-moment bounds on the particle system over finite time intervals, which follow from Proposition 3.6, we obtain

$$\mathbb{E}\|S_i\|^2 \leq C_T N^2$$

with C_T independent of N. Combining the last three displays yields

$$\mathbb{E}\|F_i^N - F[\mu_t^N](X_i, V_i)\|^2 \leq \frac{C_T}{N^2}.$$

Averaging over i proves the claim. $\square$

Lemma 7.2 (Monte Carlo fluctuation of the mean-field force)**.** *Let $(\bar{X}_i(t), \bar{V}_i(t))_{i=1}^N$ be i.i.d. copies of the nonlinear McKean–Vlasov process with common law f_t, and let*

$$\bar{\mu}_t^N := \frac{1}{N}\sum_{j=1}^{N} \delta_{(\bar{X}_j(t), \bar{V}_j(t))}$$

be the associated empirical measure. Then for every finite horizon $T > 0$ there exists $C_T < \infty$, independent of N, such that

$$\sup_{t\in[0,T]} \frac{1}{N}\sum_{i=1}^{N} \mathbb{E}\Big[\big\|F[\bar{\mu}_t^N](\bar{X}_i(t), \bar{V}_i(t)) - F[f_t](\bar{X}_i(t), \bar{V}_i(t))\big\|^2\Big] \leq \frac{C_T}{N}.$$

Proof. By exchangeability it suffices to estimate the index $i = 1$. Fix $t \in [0, T]$ and condition on $(\bar{X}_1, \bar{V}_1) = (x, v)$. Define

$$N_N(x,v) := \frac{1}{N}\sum_{j=1}^{N} K(x,\bar{X}_j)(\bar{V}_j - v), \qquad D_N(x) := \frac{1}{N}\sum_{j=1}^{N} K(x,\bar{X}_j),$$

and also

$$n_t(x,v) := \iint K(x,y)(q-v)\, f_t(dy,dq), \qquad m_t(x) := \int K(x,y)\,\rho_{f_t}(dy).$$

Then

$$F[\bar{\mu}_t^N](x,v) = \nu_{\mathrm{visc}}\frac{N_N(x,v)}{D_N(x)}, \qquad F[f_t](x,v) = \nu_{\mathrm{visc}}\frac{n_t(x,v)}{m_t(x)}.$$

Because $D_N(x) \geq k_*$ and $m_t(x) \geq k_*$,

$$\left\| \frac{N_N}{D_N} - \frac{n_t}{m_t} \right\| \leq \frac{\|N_N - n_t\|}{k_*} + \frac{\|n_t\|}{k_*^2}|D_N - m_t|.$$

Conditionally on $(\bar{X}_1, \bar{V}_1) = (x, v)$, the variables

$$Y_j := K(x,\bar{X}_j)(\bar{V}_j - v), \qquad Z_j := K(x,\bar{X}_j)$$

are independent for $j \geq 2$. Their conditional second moments satisfy

$$\mathbb{E}[\|Y_j\|^2 \mid \bar{X}_1 = x, \bar{V}_1 = v] \leq 2(k^*)^2\big(\mathbb{E}\|\bar{V}_j\|^2 + \|v\|^2\big), \qquad \mathbb{E}[|Z_j|^2 \mid \bar{X}_1 = x, \bar{V}_1 = v] \leq (k^*)^2.$$

The $j = 1$ contributions to N_N and D_N are deterministic once (x, v) is fixed and are of size $O(N^{-1}(1 + \|v\|))$, so they can be absorbed into the same N^{-1} bounds. Therefore the empirical averages have conditional variances

$$\mathbb{E}\big[\|N_N(x,v) - n_t(x,v)\|^2 \mid \bar{X}_1 = x, \bar{V}_1 = v\big] \leq \frac{C_T(1 + \|v\|^2)}{N},$$

and

$$\mathbb{E}\big[|D_N(x) - m_t(x)|^2 \mid \bar{X}_1 = x, \bar{V}_1 = v\big] \leq \frac{C_T}{N}.$$

Moreover,

$$\|n_t(x,v)\| \leq k^* \int (\|q\| + \|v\|)\, f_t(dy,dq) \leq C_T(1 + \|v\|).$$

Combining the last three displays gives

$$\mathbb{E}\Big[\|F[\bar{\mu}_t^N](\bar{X}_1,\bar{V}_1) - F[f_t](\bar{X}_1,\bar{V}_1)\|^2 \mid \bar{X}_1, \bar{V}_1\Big] \leq \frac{C_T(1 + \|\bar{V}_1\|^2)}{N}.$$

Integrating over $(\bar{X}_1, \bar{V}_1)$ and using Proposition 3.6 yields the claim. □

Theorem 7.3 (Strong propagation of chaos)**.** *Let $(\bar{X}_i, \bar{V}_i)_{i=1}^N$ be i.i.d. solutions of the nonlinear McKean–Vlasov SDE with initial law f_0, driven by the same Brownian motions as the particle system* (1)–(2)*, and started from the same initial data:*

$$(X_i(0), V_i(0)) = (\bar{X}_i(0), \bar{V}_i(0)) \qquad \textit{a.s. for every } i.$$

Set

$$\varepsilon_N(t) := \frac{1}{N}\sum_{i=1}^{N}\mathbb{E}\Big[\|X_i(t)-\bar{X}_i(t)\|^2+\|V_i(t)-\bar{V}_i(t)\|^2\Big].$$

Then for every finite horizon $T>0$ there exists $C_T<\infty$, independent of N, such that

$$\sup_{t\in[0,T]}\varepsilon_N(t)\le\frac{C_T}{N}.$$

Proof. Write

$$e_i^x := X_i-\bar{X}_i, \qquad e_i^v := V_i-\bar{V}_i, \qquad \Delta_i := F_i^N(X,V)-F[f_t](\bar{X}_i,\bar{V}_i).$$

Then

$$de_i^x = e_i^v\,dt, \qquad de_i^v = \big(-\nabla U(X_i)+\nabla U(\bar{X}_i)-\gamma e_i^v+\Delta_i\big)\,dt.$$

Itô's formula yields

$$\frac{d}{dt}\mathbb{E}\|e_i^x\|^2 = 2\mathbb{E}\langle e_i^x, e_i^v\rangle \le \mathbb{E}\|e_i^x\|^2+\mathbb{E}\|e_i^v\|^2,$$

and, by the Lipschitz bound on ∇U,

$$\begin{aligned}\frac{d}{dt}\mathbb{E}\|e_i^v\|^2 &= 2\mathbb{E}\langle e_i^v, -\nabla U(X_i)+\nabla U(\bar{X}_i)-\gamma e_i^v+\Delta_i\rangle\\ &\le (L_U+1)\mathbb{E}\|e_i^x\|^2+(L_U+1-2\gamma)\mathbb{E}\|e_i^v\|^2+\mathbb{E}\|\Delta_i\|^2.\end{aligned}$$

Adding the two inequalities and averaging over i gives

$$\frac{d}{dt}\varepsilon_N(t)\le C_0\varepsilon_N(t)+\frac{1}{N}\sum_{i=1}^{N}\mathbb{E}\|\Delta_i(t)\|^2 \tag{27}$$

for a constant C_0 depending only on (L_U,γ).

Decompose

$$\Delta_i = R_i+S_i+T_i,$$

where

$$\begin{aligned}R_i &:= F_i^N(X,V)-F[\mu_t^N](X_i,V_i),\\ S_i &:= F[\mu_t^N](X_i,V_i)-F[\bar{\mu}_t^N](\bar{X}_i,\bar{V}_i),\\ T_i &:= F[\bar{\mu}_t^N](\bar{X}_i,\bar{V}_i)-F[f_t](\bar{X}_i,\bar{V}_i).\end{aligned}$$

Using $(a+b+c)^2\le 3(a^2+b^2+c^2)$,

$$\frac{1}{N}\sum_{i=1}^{N}\mathbb{E}\|\Delta_i\|^2\le 3\left(\frac{1}{N}\sum_i\mathbb{E}\|R_i\|^2+\frac{1}{N}\sum_i\mathbb{E}\|S_i\|^2+\frac{1}{N}\sum_i\mathbb{E}\|T_i\|^2\right).$$

Lemma 7.1 yields

$$\frac{1}{N}\sum_i\mathbb{E}\|R_i\|^2\le\frac{C_T}{N},$$

and Lemma 7.2 yields

$$\frac{1}{N}\sum_i\mathbb{E}\|T_i\|^2\le\frac{C_T}{N}.$$

For the middle term, Lemma 3.4 gives

$$\|S_i\| \le L_{\mathrm{mf}}\Big(\|e_i^x\| + \|e_i^v\| + W_1(\mu_t^N, \bar{\mu}_t^N)\Big).$$

Coupling the empirical measures particlewise yields

$$W_1(\mu_t^N, \bar{\mu}_t^N) \le \frac{1}{N}\sum_{j=1}^N \big(\|e_j^x\| + \|e_j^v\|\big),$$

and Jensen's inequality therefore gives

$$\mathbb{E}W_1(\mu_t^N, \bar{\mu}_t^N)^2 \le 2\varepsilon_N(t).$$

Consequently

$$\frac{1}{N}\sum_{i=1}^N \mathbb{E}\|S_i\|^2 \le C_1\varepsilon_N(t).$$

Substituting into (27) yields

$$\frac{d}{dt}\varepsilon_N(t) \le C_2\varepsilon_N(t) + \frac{C_T}{N}.$$

Because $\varepsilon_N(0) = 0$, Grönwall's lemma gives

$$\sup_{t\in[0,T]} \varepsilon_N(t) \le \frac{C_T}{N}.$$

□

Theorem 7.4 (Relative entropy on path space and at fixed time)**.** *Under the hypotheses of Theorem 7.3, let $P^N_{[0,T]}$ be the law on path space $C([0,T];(\mathbb{R}^{2d})^N)$ of the interacting system (1)–(2), and let $Q^N_{[0,T]}$ be the law of the independent nonlinear system from Theorem 7.3. Then for every finite horizon $T > 0$ there exists $C_T < \infty$, independent of N, such that*

$$\mathrm{KL}\big(P^N_{[0,T]} \,\|\, Q^N_{[0,T]}\big) \le C_T. \tag{28}$$

In particular, for every $t \in [0,T]$,

$$\mathrm{KL}\big(\rho_t^N \,\|\, f_t^{\otimes N}\big) \le C_T. \tag{29}$$

Proof. Under $Q^N_{[0,T]}$, the processes satisfy

$$d\bar{V}_i = \big(-\nabla U(\bar{X}_i) - \gamma\bar{V}_i + F[f_t](\bar{X}_i, \bar{V}_i)\big)\,dt + \sigma\,dW_i.$$

The interacting system differs only in the velocity drift. Since the diffusion matrix is constant in the velocity variables, Girsanov's theorem applies. The Radon–Nikodým derivative of $P^N_{[0,T]}$ with respect to $Q^N_{[0,T]}$ is

$$\frac{dP^N_{[0,T]}}{dQ^N_{[0,T]}} = \exp\left(\frac{1}{\sigma}\sum_{i=1}^N \int_0^T \Delta_i(s)\cdot dW_i(s) - \frac{1}{2\sigma^2}\sum_{i=1}^N \int_0^T \|\Delta_i(s)\|^2\,ds\right),$$

with

$$\Delta_i(s) := F_i^N(X_s, V_s) - F[f_s](X_i(s), V_i(s)).$$

The moment bounds from Proposition 3.6, together with Lemma 7.1 and Theorem 7.3, imply Novikov's criterion on every finite interval. Taking expectations under $P^N_{[0,T]}$ gives the exact entropy identity

$$\mathrm{KL}\big(P^N_{[0,T]} \,\|\, Q^N_{[0,T]}\big) = \frac{1}{2\sigma^2} \sum_{i=1}^{N} \mathbb{E} \int_0^T \|\Delta_i(s)\|^2 \, ds.$$

We estimate the drift discrepancy by inserting the coupled empirical measure $\bar{\mu}^N_t$ from Theorem 7.3. Write

$$\Delta_i = R_i + U_i + T_i + V_i,$$

where

$$\begin{aligned} R_i &:= F^N_i(X, V) - F[\mu^N_t](X_i, V_i), \\ U_i &:= F[\mu^N_t](X_i, V_i) - F[\bar{\mu}^N_t](\bar{X}_i, \bar{V}_i), \\ T_i &:= F[\bar{\mu}^N_t](\bar{X}_i, \bar{V}_i) - F[f_t](\bar{X}_i, \bar{V}_i), \\ V_i &:= F[f_t](\bar{X}_i, \bar{V}_i) - F[f_t](X_i, V_i). \end{aligned}$$

Then $(a + b + c + d)^2 \leq 4(a^2 + b^2 + c^2 + d^2)$ gives

$$\frac{1}{N} \sum_{i=1}^{N} \mathbb{E}\|\Delta_i(s)\|^2 \leq \frac{4}{N} \sum_{i=1}^{N} \mathbb{E}\|R_i(s)\|^2 + \frac{4}{N} \sum_{i=1}^{N} \mathbb{E}\|U_i(s)\|^2 + \frac{4}{N} \sum_{i=1}^{N} \mathbb{E}\|T_i(s)\|^2 + \frac{4}{N} \sum_{i=1}^{N} \mathbb{E}\|V_i(s)\|^2.$$

Lemma 7.1 controls the R_i term, and Lemma 7.2 controls the T_i term:

$$\frac{1}{N} \sum_i \mathbb{E}\|R_i(s)\|^2 \leq \frac{C_T}{N}, \qquad \frac{1}{N} \sum_i \mathbb{E}\|T_i(s)\|^2 \leq \frac{C_T}{N}.$$

For the U_i term, Lemma 3.4 gives

$$\|U_i\| \leq L_{\mathrm{mf}} \Big(\|X_i - \bar{X}_i\| + \|V_i - \bar{V}_i\| + W_1(\mu^N_t, \bar{\mu}^N_t) \Big).$$

Exactly as in the proof of Theorem 7.3,

$$\mathbb{E} W_1(\mu^N_t, \bar{\mu}^N_t)^2 \leq 2\varepsilon_N(t),$$

and therefore

$$\frac{1}{N} \sum_i \mathbb{E}\|U_i(s)\|^2 \leq C_T \varepsilon_N(s).$$

For the last term, the Lipschitz bound of Lemma 3.4 gives

$$\frac{1}{N} \sum_i \mathbb{E}\|V_i(s)\|^2 \leq C_T \varepsilon_N(s).$$

Combining the last four displays and using Theorem 7.3, we obtain

$$\frac{1}{N} \sum_{i=1}^{N} \mathbb{E}\|\Delta_i(s)\|^2 \leq C_T \varepsilon_N(s) + \frac{C_T}{N} \leq \frac{C_T}{N}$$

for all $s \in [0, T]$. Integrating in time gives

$$\sum_{i=1}^{N} \mathbb{E} \int_0^T \|\Delta_i(s)\|^2 \, ds \leq C_T,$$

which proves (28). The fixed-time bound (29) follows by data processing, since evaluation at time t is a measurable map from path space to $(\mathbb{R}^{2d})^N$. □

Corollary 7.5 (One-particle total-variation error). *Let $\rho_{t,1}^N$ be the first marginal of ρ_t^N. Under the assumptions of Theorem 7.4,*

$$\|\rho_{t,1}^N - f_t\|_{\mathrm{TV}} \le \sqrt{\frac{2C_T}{N}} \qquad \forall t \in [0,T].$$

Proof. For any probability measure P^N on $(\mathbb{R}^{2d})^N$ and any product measure $Q^{\otimes N}$, the chain rule for relative entropy gives

$$\mathrm{KL}(P^N \,\|\, Q^{\otimes N}) = \sum_{i=1}^{N} \mathbb{E}_{P^N}\Big[\mathrm{KL}\big(P^N_{i\,|\,1:i-1}(\cdot \mid Z_1,\dots,Z_{i-1}) \,\|\, Q\big)\Big].$$

Since $\nu \mapsto \mathrm{KL}(\nu\|Q)$ is convex,

$$\mathbb{E}_{P^N}\Big[\mathrm{KL}\big(P^N_{i\,|\,1:i-1}(\cdot \mid Z_1,\dots,Z_{i-1}) \,\|\, Q\big)\Big] \ge \mathrm{KL}(P_i^N \,\|\, Q).$$

Applying this to $P^N = \rho_t^N$ and $Q = f_t$, and using exchangeability of ρ_t^N, we obtain

$$\mathrm{KL}(\rho_{t,1}^N \,\|\, f_t) \le \frac{1}{N}\mathrm{KL}(\rho_t^N \,\|\, f_t^{\otimes N}).$$

Now use (29) and Pinsker's inequality. □

8 Stationary Mean-Field Limit and Mean-Field LSI

8.1 Exchangeability and finite de Finetti reduction

Theorem 8.1 (Exchangeability of the invariant law). *For every $N \ge 2$, the invariant law π_N of the finite-particle kinetic core is exchangeable:*

$$\sigma_{\#}\pi_N = \pi_N \qquad \forall \sigma \in S_N,$$

where σ acts on $(\mathbb{R}^{2d})^N$ by permutation of particle labels.

Proof. Fix a permutation $\sigma \in S_N$ and let

$$\Sigma_\sigma(x_1, v_1, \dots, x_N, v_N) := (x_{\sigma(1)}, v_{\sigma(1)}, \dots, x_{\sigma(N)}, v_{\sigma(N)}).$$

Because the confining force $-\nabla U(x_i)$, the friction $-\gamma v_i$, the velocity noise $\sigma\, dW_i$, and the row-normalized viscous force

$$F_i(X,V) = \nu_{\mathrm{visc}} \sum_{j \ne i} \omega_{ij}(X)(v_j - v_i)$$

are defined by the same formula for every label i, the generator $\mathcal{L}_N$ of (1)–(2) commutes with every permutation:

$$\mathcal{L}_N(\varphi \circ \Sigma_\sigma) = (\mathcal{L}_N \varphi) \circ \Sigma_\sigma \qquad \forall \varphi \in C_c^\infty((\mathbb{R}^{2d})^N).$$

Equivalently, if P_t^N is the Markov semigroup of the particle system, then

$$P_t^N(\varphi \circ \Sigma_\sigma) = (P_t^N \varphi) \circ \Sigma_\sigma.$$

If π_N is invariant, then for every bounded measurable φ,

$$\int \varphi \, d(\sigma_{\#}\pi_N) = \int \varphi \circ \Sigma_\sigma \, d\pi_N$$

$$
\begin{aligned}
&= \int P_t^N(\varphi \circ \Sigma_\sigma)\, d\pi_N \\
&= \int (P_t^N \varphi) \circ \Sigma_\sigma\, d\pi_N \\
&= \int P_t^N \varphi\, d(\sigma_\# \pi_N).
\end{aligned}
$$

Thus $\sigma_\# \pi_N$ is another invariant probability measure for the same semigroup. By the uniqueness of π_N from Theorem 5.4, $\sigma_\# \pi_N = \pi_N$. □

Theorem 8.2 (Finite de Finetti representation)**.** *For each $N \geq 2$ there exists a probability measure Q_N on $\mathcal{P}(\mathbb{R}^{2d})$ such that, for every $1 \leq k \leq N$,*

$$
d_{\mathrm{TV}}\left(\pi_{N,k}, \int_{\mathcal{P}(\mathbb{R}^{2d})} \mu^{\otimes k}\, Q_N(d\mu)\right) \leq \frac{k(k-1)}{2N},
$$

where $\pi_{N,k}$ is the law of the first k particles under π_N. One may take Q_N to be the law of the empirical measure

$$
L_N := \frac{1}{N} \sum_{i=1}^{N} \delta_{(X_i, V_i)} \qquad \text{under } \pi_N.
$$

Proof. The state space $\mathbb{R}^{2d}$ is a standard Borel space, and Theorem 8.1 shows that π_N is exchangeable on $(\mathbb{R}^{2d})^N$. The finite de Finetti theorem of Diaconis and Freedman [2] therefore applies directly. It gives a probability measure Q_N on $\mathcal{P}(\mathbb{R}^{2d})$ such that

$$
d_{\mathrm{TV}}\left(\pi_{N,k}, \int \mu^{\otimes k}\, Q_N(d\mu)\right) \leq \frac{k(k-1)}{2N} \qquad \forall\, 1 \leq k \leq N.
$$

For the canonical choice, define the empirical-measure map

$$
\mathfrak{L}_N : (\mathbb{R}^{2d})^N \to \mathcal{P}(\mathbb{R}^{2d}), \qquad \mathfrak{L}_N(z_1, \ldots, z_N) := \frac{1}{N} \sum_{i=1}^{N} \delta_{z_i},
$$

and set $Q_N := (\mathfrak{L}_N)_\# \pi_N$. This is exactly the law of L_N under π_N, and the same Diaconis–Freedman theorem gives the stated bound for this canonical mixing measure. □

Corollary 8.3 (Exact barycentric representation of the first marginal)**.** *If $\pi_{N,1}$ denotes the first marginal of π_N, then*

$$
\pi_{N,1} = \int_{\mathcal{P}(\mathbb{R}^{2d})} \mu\, Q_N(d\mu).
$$

Proof. Apply Theorem 8.2 with $k = 1$. Since

$$
\frac{1(1-1)}{2N} = 0,
$$

the total-variation distance is zero:

$$
d_{\mathrm{TV}}\left(\pi_{N,1}, \int \mu\, Q_N(d\mu)\right) = 0.
$$

Two probability measures at total-variation distance zero are equal, so the first marginal is exactly the barycenter of Q_N. □

Lemma 8.4 (Conditioning on the empirical measure)**.** *Let $G : \mathbb{R}^{2d} \times \mathcal{P}(\mathbb{R}^{2d}) \to \mathbb{R}$ be bounded and measurable. Under the exchangeable law π_N,*

$$\mathbb{E}_{\pi_N}[G((X_1, V_1), L_N)] = \mathbb{E}_{\pi_N}\left[\int_{\mathbb{R}^{2d}} G(z, L_N)\, L_N(dz)\right].$$

Proof. By exchangeability, each index i has the same law as index 1, so

$$\mathbb{E}_{\pi_N}[G((X_1, V_1), L_N)] = \frac{1}{N}\sum_{i=1}^{N} \mathbb{E}_{\pi_N}[G((X_i, V_i), L_N)].$$

Inside the expectation, the empirical average is exactly the integral against the empirical measure:

$$\frac{1}{N}\sum_{i=1}^{N} G((X_i, V_i), L_N) = \int_{\mathbb{R}^{2d}} G(z, L_N)\, L_N(dz).$$

Taking expectations of both sides gives the claimed identity. □

8.2 Tightness, identification, and the stationary thermodynamic limit

Theorem 8.5 (Tightness of the stationary marginals)**.** *The sequence of first marginals $\{\pi_{N,1}\}_{N\geq 2}$ is tight in $\mathcal{P}(\mathbb{R}^{2d})$.*

Proof. The finite-particle ergodic construction underlying the invariant measures π_N provides an N-uniform Foster–Lyapunov bound for a quadratic phase function. Concretely, there exists a function $V_N : (\mathbb{R}^{2d})^N \to [1, \infty)$ of the form

$$V_N(X, V) \asymp 1 + \frac{1}{N}\sum_{i=1}^{N}\big(\|X_i\|^2 + \|V_i\|^2\big),$$

and constants $\lambda_V > 0$, $C_V < \infty$, independent of N, such that

$$\mathcal{L}_N V_N \leq -\lambda_V V_N + C_V.$$

Integrating against the invariant law π_N and using $\int \mathcal{L}_N V_N\, d\pi_N = 0$ yields

$$\int V_N\, d\pi_N \leq \frac{C_V}{\lambda_V}.$$

By the equivalence built into the definition of V_N, there exists $C_2 < \infty$, independent of N, such that

$$\int \frac{1}{N}\sum_{i=1}^{N}\big(\|x_i\|^2 + \|v_i\|^2\big)\, \pi_N(dX\, dV) \leq C_2.$$

Using exchangeability from Theorem 8.1, this becomes

$$\int_{\mathbb{R}^{2d}} (\|x\|^2 + \|v\|^2)\, \pi_{N,1}(dx\, dv) \leq C_2 \qquad \forall N \geq 2.$$

Now fix $\varepsilon > 0$ and choose $R_\varepsilon := \sqrt{C_2/\varepsilon}$. Markov's inequality gives

$$\pi_{N,1}\big(\{(x, v) : \|x\|^2 + \|v\|^2 > R_\varepsilon^2\}\big) \leq \frac{C_2}{R_\varepsilon^2} = \varepsilon.$$

Hence the compact ball $K_\varepsilon := \{(x, v) : \|x\|^2 + \|v\|^2 \leq R_\varepsilon^2\}$ satisfies

$$\pi_{N,1}(K_\varepsilon) \geq 1 - \varepsilon \qquad \forall N \geq 2.$$

Prokhorov's theorem therefore yields tightness. □

Proposition 8.6 (Unique stationary mean-field law)**.** *There exists a unique stationary probability law $\pi_\infty \in \mathcal{P}(\mathbb{R}^{2d})$ for the mean-field equation* (5)*. Equivalently, π_∞ is the unique probability law such that*

$$\int_{\mathbb{R}^{2d}} \mathcal{L}_{\pi_\infty}\phi(z)\,\pi_\infty(dz) = 0 \qquad \forall \phi \in C_c^\infty(\mathbb{R}^{2d}), \tag{30}$$

where

$$\mathcal{L}_\mu \phi(x,v) := v\cdot\nabla_x\phi + \big(-\nabla U(x) - \gamma v + F[\mu](x,v)\big)\cdot\nabla_v\phi + \frac{\sigma^2}{2}\Delta_v\phi.$$

Proof. We use the fixed-point argument based on the weighted resolvent of the kinetic Fokker–Planck operator. Let

$$w(x,v) := 1 + \|x\|^2 + \|v\|^2, \qquad H_w^1 := \left\{\rho : \int (\rho^2 + \|\nabla_x\rho\|^2 + \|\nabla_v\rho\|^2) w < \infty\right\}.$$

For a radius $R > 0$ fixed by the a priori resolvent estimate, set

$$\mathcal{P}_R := \Big\{\rho \in H_w^1 : \rho \geq 0,\ \int_{\mathbb{R}^{2d}} \rho = 1,\ \|\rho\|_{H_w^1} \leq R\Big\}.$$

Then $\mathcal{P}_R$ is a closed subset of the Banach space H_w^1, hence complete. The weighted norm on $\mathcal{P}_R$ gives a uniform first-moment bound on that ball, so Lemma 3.4 applies there with one constant throughout the argument.

Write the linear kinetic adjoint as

$$\mathcal{L}_0^*\rho = -\nabla_x\cdot(v\rho) + \nabla_v\cdot((\nabla U(x) + \gamma v)\rho) + \frac{\sigma^2}{2}\Delta_v\rho.$$

Choose $C > 0$ and rewrite the stationary equation $\mathcal{L}_0^*\rho - \nabla_v\cdot(F[\rho]\rho) = 0$ as

$$(C - \mathcal{L}_0^*)\rho = C\rho - \nabla_v\cdot(F[\rho]\rho).$$

Define the solution operator

$$\mathcal{T}[\rho] := (C - \mathcal{L}_0^*)^{-1}\big(C\rho - \nabla_v\cdot(F[\rho]\rho)\big).$$

Standard weighted resolvent theory for the underdamped Fokker–Planck operator with uniformly convex potential yields a bounded map

$$(C - \mathcal{L}_0^*)^{-1} : L_w^2 \to H_w^1,$$

whose operator norm we denote by M_C and which satisfies $M_C \to 0$ as $C \to \infty$. The positivity-preserving property of the kinetic semigroup implies $\mathcal{T}[\rho] \geq 0$ whenever $\rho \geq 0$, and integrating the defining equation over $\mathbb{R}^{2d}$ shows

$$\int_{\mathbb{R}^{2d}} \mathcal{T}[\rho] = 1 \qquad \text{whenever} \qquad \int_{\mathbb{R}^{2d}} \rho = 1,$$

because both the transport divergence and the velocity divergence have zero integral. Thus $\mathcal{T}$ maps the probability-density cone of H_w^1 into itself.

Next we estimate the non-linear part. Since

$$F[\rho](x,v) = \nu_{\text{visc}}\big(A[\rho](x) - v\big)$$

is affine in v, the divergence difference splits as

$$\nabla_v \cdot (F[\rho_1]\rho_1 - F[\rho_2]\rho_2) = \nabla_v \cdot \big((F[\rho_1] - F[\rho_2])\rho_1\big) + \nabla_v \cdot \big(F[\rho_2](\rho_1 - \rho_2)\big).$$

The uniform first-moment bound on $\mathcal{P}_R$ and Lemma 3.4 therefore give a finite constant $L_{\mathrm{nl}}(R)$ such that

$$\big\|\nabla_v \cdot (F[\rho_1]\rho_1 - F[\rho_2]\rho_2)\big\|_{L^2_w} \le L_{\mathrm{nl}}(R)\|\rho_1 - \rho_2\|_{H^1_w} \qquad \forall \rho_1, \rho_2 \in \mathcal{P}_R.$$

Applying the resolvent bound therefore gives

$$\|\mathcal{T}[\rho_1] - \mathcal{T}[\rho_2]\|_{H^1_w} \le M_C\big(C + L_{\mathrm{nl}}(R)\big)\|\rho_1 - \rho_2\|_{H^1_w}.$$

Because $M_C \to 0$ as $C \to \infty$, we may choose C so large that

$$\kappa_R := M_C\big(C + L_{\mathrm{nl}}(R)\big) < 1.$$

The a priori resolvent estimate used to choose R also gives $\mathcal{T}[\mathcal{P}_R] \subseteq \mathcal{P}_R$, so $\mathcal{T}$ is a strict contraction on the complete metric space $\mathcal{P}_R$. Banach's fixed-point theorem yields a unique fixed point, and the fixed-point equation is exactly (30). This fixed point is π_∞. □

Theorem 8.7 (Stationary propagation of chaos)**.** *The sequence of first marginals $\pi_{N,1}$ converges weakly to the unique stationary mean-field law π_∞ from Proposition 8.6. Equivalently,*

$$\pi_{N,1} \Rightarrow \pi_\infty \qquad \text{as } N \to \infty.$$

Proof. We now carry out the tightness–identification–uniqueness program in full.

Step 1: tightness. By Theorem 8.5, the family $\{\pi_{N,1}\}_{N\ge2}$ is tight. Hence every subsequence admits a weakly convergent sub-subsequence.

Step 2: stationarity identity for the finite-N system. Fix $\phi \in C_c^\infty(\mathbb{R}^{2d})$ and set

$$\Phi(X_1, V_1, \ldots, X_N, V_N) := \phi(X_1, V_1).$$

Since π_N is invariant,

$$\int \mathcal{L}_N \Phi \, d\pi_N = 0.$$

Writing

$$\mathcal{L}_\mu\phi(x,v) := v \cdot \nabla_x\phi + \big(-\nabla U(x) - \gamma v + F[\mu](x,v)\big) \cdot \nabla_v\phi + \frac{\sigma^2}{2}\Delta_v\phi,$$

this gives

$$\int \mathcal{L}_0\phi(z_1)\,\pi_N(dZ) + \int F_1^N(Z) \cdot \nabla_v\phi(z_1)\,\pi_N(dZ) = 0,$$

where $\mathcal{L}_0$ is the backbone generator without the mean-field term. By Lemma 7.1, evaluated under the stationary law π_N,

$$\left|\int \big(F_1^N - F[L_N](z_1)\big) \cdot \nabla_v\phi(z_1)\,d\pi_N\right| \le \|\nabla_v\phi\|_\infty \left(\int \|F_1^N - F[L_N](z_1)\|^2\,d\pi_N\right)^{1/2} \le \frac{C_\phi}{\sqrt{N}}.$$

$$\Psi_\phi(\mu) := \int_{\mathbb{R}^{2d}} \mathcal{L}_\mu\phi(z)\,\mu(dz).$$

By Corollary 8.3 and Lemma 8.4,

$$|\mathbb{E}_{\pi_N}[\Psi_\phi(L_N)]| \le \frac{C_\phi}{\sqrt{N}}. \tag{31}$$

Step 4: identification of limit points. Let $\{N_k\}$ be a subsequence such that $\pi_{N_k,1} \Rightarrow \bar{\pi}$. By the uniform second-moment bound in Theorem 8.5 and the canonical choice $Q_N = \mathrm{Law}(L_N)$ from Theorem 8.2, the family $\{Q_N\}_{N\ge 2}$ is tight in $\mathcal{P}(\mathcal{P}(\mathbb{R}^{2d}))$. Passing to a further subsequence if necessary, we may assume $Q_{N_k} \Rightarrow Q_\infty$. Indeed,

$$\int_{\mathcal{P}(\mathbb{R}^{2d})} \left[\int_{\mathbb{R}^{2d}} (\|x\|^2 + \|v\|^2)\, \mu(dx\, dv)\right] Q_N(d\mu) = \int_{\mathbb{R}^{2d}} (\|x\|^2 + \|v\|^2)\, \pi_{N,1}(dx\, dv) \le C_2,$$

so the same Markov–Prokhorov argument applies on the second level. For compactly supported ϕ, the map Ψ_ϕ is continuous on $\mathcal{P}(\mathbb{R}^{2d})$ under weak convergence together with uniform second moments, because K and $\nabla_v\phi$ are bounded and the denominator in $F[\mu]$ is uniformly bounded below by k_*. Passing to the limit in the previous display therefore gives

$$\int_{\mathcal{P}(\mathbb{R}^{2d})} \Psi_\phi(\mu)\, Q_\infty(d\mu) = 0 \qquad \forall \phi \in C_c^\infty(\mathbb{R}^{2d}).$$

This is the stationary weak mean-field identity for the limit mixture. The vanishing defect estimate (31), together with the continuity just noted, puts us in the hypotheses of the standard nonlinear martingale-problem identification for exchangeable particle systems (equivalently, the de Finetti/Sznitman limit-point argument; see [6]). Hence

$$Q_\infty\Big(\big\{\mu : \int_{\mathbb{R}^{2d}} \mathcal{L}_\mu \phi(z)\, \mu(dz) = 0 \;\forall \phi \in C_c^\infty(\mathbb{R}^{2d})\big\}\Big) = 1.$$

By Proposition 8.6, this set consists of the single law π_∞, so $Q_\infty = \delta_{\pi_\infty}$.

Step 5: uniqueness and collapse of the mixing measure. Since the barycenter map is continuous on the uniform second-moment class,

$$\bar{\pi} = \int \mu\, Q_\infty(d\mu) = \int \mu\, \delta_{\pi_\infty}(d\mu) = \pi_\infty.$$

Thus every convergent subsequence of $\{\pi_{N,1}\}_{N\ge 2}$ has the same limit, so the whole sequence converges:

$$\pi_{N,1} \Rightarrow \pi_\infty.$$

□

Corollary 8.8 (Thermodynamic limit for bounded observables)**.** *For every bounded continuous observable* $\phi : \mathbb{R}^{2d} \to \mathbb{R}$*,*

$$\lim_{N\to\infty} \mathbb{E}_{\pi_N}\left[\frac{1}{N}\sum_{i=1}^N \phi(X_i, V_i)\right] = \int_{\mathbb{R}^{2d}} \phi\, d\pi_\infty.$$

Proof. Exchangeability gives

$$\mathbb{E}_{\pi_N}\left[\frac{1}{N}\sum_{i=1}^N \phi(X_i, V_i)\right] = \int_{\mathbb{R}^{2d}} \phi\, d\pi_{N,1}.$$

Now apply Theorem 8.7 and the definition of weak convergence. □

Theorem 8.9 (Mean-field logarithmic Sobolev inequality). *Let*

$$\Gamma^{\infty}_{\text{hypo}}(g, g) := a\|\nabla_v g\|^2 + 2b\, \nabla_x g \cdot \nabla_v g + c\|\nabla_x g\|^2, \qquad ac > b^2,$$

with the same coefficients a, b, c as in Theorem 5.4. Then the stationary mean-field law π_∞ satisfies

$$\mathrm{Ent}_{\pi_\infty}(g^2) \le C_{\mathrm{LSI}} \int_{\mathbb{R}^{2d}} \Gamma^{\infty}_{\text{hypo}}(g, g)\ \mathrm{d}\pi_\infty$$

for every smooth compactly supported g, with the same N-uniform constant C_{LSI} as in (25).

Proof. Fix $g \in C_c^\infty(\mathbb{R}^{2d})$ and extend it to $(\mathbb{R}^{2d})^N$ by

$$G_N(z_1, \ldots, z_N) := g(z_1).$$

Applying the finite-particle inequality (25) to G_N gives

$$\mathrm{Ent}_{\pi_N}(G_N^2) \le C_{\mathrm{LSI}} \int \Gamma^{N}_{\text{hypo}}(G_N, G_N)\, d\pi_N.$$

Because G_N depends only on the first particle,

$$\mathrm{Ent}_{\pi_N}(G_N^2) = \mathrm{Ent}_{\pi_{N,1}}(g^2), \qquad \Gamma^{N}_{\text{hypo}}(G_N, G_N)(z_1, \ldots, z_N) = \Gamma^{\infty}_{\text{hypo}}(g, g)(z_1).$$

Therefore

$$\mathrm{Ent}_{\pi_{N,1}}(g^2) \le C_{\mathrm{LSI}} \int_{\mathbb{R}^{2d}} \Gamma^{\infty}_{\text{hypo}}(g, g)\, d\pi_{N,1}.$$

Theorem 8.7 gives $\pi_{N,1} \Rightarrow \pi_\infty$. Since g is smooth and compactly supported, the functions g^2, $g^2 \log g^2$, and $\Gamma^{\infty}_{\text{hypo}}(g, g)$ are bounded and continuous on $\mathbb{R}^{2d}$ (with the convention $0 \log 0 = 0$). Weak convergence therefore implies

$$\mathrm{Ent}_{\pi_{N,1}}(g^2) \to \mathrm{Ent}_{\pi_\infty}(g^2), \qquad \int \Gamma^{\infty}_{\text{hypo}}(g, g)\, d\pi_{N,1} \to \int \Gamma^{\infty}_{\text{hypo}}(g, g)\, d\pi_\infty.$$

Passing to the limit in the finite-N inequality yields

$$\mathrm{Ent}_{\pi_\infty}(g^2) \le C_{\mathrm{LSI}} \int \Gamma^{\infty}_{\text{hypo}}(g, g)\, d\pi_\infty,$$

which is the desired mean-field logarithmic Sobolev inequality. $\square$

8.3 Translated confinement and the wall-to-infinity limit

Definition 8.10 (Translated confining family). Fix a reference potential $U_* : \mathbb{R}^d \to \mathbb{R}$ satisfying Assumption 2.1. For each translation vector $a \in \mathbb{R}^d$, define

$$U_a(x) := U_*(x - a).$$

Let $P_t^{N,a}$, $Q_\tau^{N,a}$, π_N^a, and π_∞^a denote, respectively, the continuous finite-particle semigroup, the squashed observation chain, the invariant law, and the stationary mean-field law for the kinetic core with confining potential U_a.

Define the translation maps

$$t_a(x, v) := (x + a, v) \in \mathbb{R}^{2d},$$

$$T_a^N(x_1, v_1, \ldots, x_N, v_N) := \big(x_1 + a, v_1, \ldots, x_N + a, v_N\big) \in (\mathbb{R}^{2d})^N.$$

Theorem 8.11 (Translation covariance of the confining family)**.** *For every $a \in \mathbb{R}^d$, the translated family from Definition 8.10 satisfies the following exact covariance relations.*

(i) *If $(X_i^0(t), V_i^0(t))_{i=1}^N$ solves* (1)–(2) *with potential U_*, then*

$$X_i^a(t) := X_i^0(t) + a, \qquad V_i^a(t) := V_i^0(t)$$

solves the same system with potential U_a, driven by the same Brownian motions.

(ii) *The continuous semigroups and squashed observation chains satisfy*

$$P_t^{N,a}(F \circ T_a^N) = (P_t^{N,0} F) \circ T_a^N, \qquad Q_\tau^{N,a}(F \circ T_a^N) = (Q_\tau^{N,0} F) \circ T_a^N$$

for every bounded measurable F.

(iii) *The invariant laws satisfy*

$$T_{a\#}^N \pi_N^0 = \pi_N^a.$$

(iv) *If f_t^0 solves the mean-field equation* (5) *with potential U_*, then*

$$f_t^a := t_{a\#} f_t^0$$

solves the translated mean-field equation with potential U_a. In particular,

$$t_{a\#}\pi_\infty^0 = \pi_\infty^a.$$

(v) *The constants in the finite-N hypocoercive logarithmic Sobolev inequality* (25)*, the entropy contraction* (26)*, and the mean-field logarithmic Sobolev inequality from Theorem 8.9 are the same for every a.*

Proof. The kernel is translation invariant:

$$K(x_i + a, x_j + a) = K(x_i, x_j).$$

Hence

$$d_i(X + a\mathbf{1}) = d_i(X), \qquad \omega_{ij}(X + a\mathbf{1}) = \omega_{ij}(X),$$

and therefore the viscous force is translation covariant:

$$F_i(X + a\mathbf{1}, V) = F_i(X, V).$$

The confining drift transforms as

$$\nabla U_a(x + a) = \nabla U_*(x).$$

Substituting these identities into (1)–(2) shows that if (X_i^0, V_i^0) solves the reference system, then $(X_i^0 + a, V_i^0)$ solves the translated system. This proves (i).

For (ii), let $Z_t^{N,0}$ be the reference process and $Z_t^{N,a} := T_a^N Z_t^{N,0}$. By (i), $Z_t^{N,a}$ has the law of the translated process started from $T_a^N z$ whenever $Z_t^{N,0}$ starts from z. Therefore

$$P_t^{N,a}(F \circ T_a^N)(z) = \mathbb{E}_z\big[F(T_a^N Z_t^{N,a})\big] = \mathbb{E}_{T_a^N z}\big[F(Z_t^{N,0})\big] = (P_t^{N,0} F)(T_a^N z).$$

The observation map $S(x, v) = (x, \psi(v))$ commutes with translations in the position variables:

$$S \circ T_a^N = T_a^N \circ S.$$

Hence the same covariance relation holds for the observed chain $Q_\tau^{N,a}$.

For (iii), let $\mu := T^N_{a\#}\pi^0_N$. Then for every bounded measurable F,

$$\int P_t^{N,a}F\,d\mu = \int P_t^{N,a}F(T_a^N z)\,\pi_N^0(dz) = \int P_t^{N,0}(F\circ T_a^N)(z)\,\pi_N^0(dz) = \int F(T_a^N z)\,\pi_N^0(dz) = \int F\,d\mu,$$

where the third equality uses invariance of π_N^0 under $P_t^{N,0}$. Thus μ is invariant for the translated dynamics. By the uniqueness part of Theorem 5.4, $\mu = \pi_N^a$.

For (iv), let f_t^0 be a weak solution of (5) with potential U_*, and define $f_t^a = t_{a\#}f_t^0$. Because K is translation invariant, the normalized kernel mass transforms by

$$m_{f_t^a}(x+a) = m_{f_t^0}(x),$$

and therefore

$$F[f_t^a](x+a,v) = F[f_t^0](x,v).$$

Writing the translated mean-field SDE with potential U_a,

$$d\bar{X}_t^a = \bar{V}_t^a\,dt, \qquad d\bar{V}_t^a = \big(-\nabla U_a(\bar{X}_t^a) - \gamma\bar{V}_t^a + F[f_t^a](\bar{X}_t^a,\bar{V}_t^a)\big)\,dt + \sigma\,dW_t,$$

and setting $(\bar{X}_t^a,\bar{V}_t^a) := (\bar{X}_t^0 + a, \bar{V}_t^0)$ gives exactly the reference SDE with potential U_*. Hence $f_t^a = t_{a\#}f_t^0$ is the translated mean-field solution. Applying this to the unique stationary law from Theorem 8.7 yields $t_{a\#}\pi_\infty^0 = \pi_\infty^a$.

For (v), the pointwise derivatives are unchanged by translation in the position variables. Consequently, for every smooth g,

$$\Gamma^N_{\mathrm{hypo}}(g\circ T_a^N, g\circ T_a^N) = \Gamma^N_{\mathrm{hypo}}(g,g)\circ T_a^N,$$

and similarly for the mean-field carré du champ $\Gamma^\infty_{\mathrm{hypo}}$. Using the pushforward identities from (iii) and (iv),

$$\mathrm{Ent}_{\pi_N^a}(g^2) = \mathrm{Ent}_{\pi_N^0}\big((g\circ T_a^N)^2\big), \qquad \int \Gamma^N_{\mathrm{hypo}}(g,g)\,d\pi_N^a = \int \Gamma^N_{\mathrm{hypo}}(g\circ T_a^N, g\circ T_a^N)\,d\pi_N^0,$$

and likewise

$$\mathrm{Ent}_{\pi_\infty^a}(g^2) = \mathrm{Ent}_{\pi_\infty^0}\big((g\circ t_a)^2\big), \qquad \int \Gamma^\infty_{\mathrm{hypo}}(g,g)\,d\pi_\infty^a = \int \Gamma^\infty_{\mathrm{hypo}}(g\circ t_a, g\circ t_a)\,d\pi_\infty^0.$$

Therefore the finite-N and mean-field logarithmic Sobolev inequalities for the translated family are identical to those of the reference family, with the same constants. The entropy contraction constants are the same because the semigroup covariance in (ii) conjugates the translated dynamics to the reference dynamics. □

Corollary 8.12 (Thermodynamic limit followed by wall-to-infinity recentering)**.** *Let $(a_R)_{R\geq 1} \subset \mathbb{R}^d$ satisfy $\|a_R\| \to \infty$. For each R, consider the translated confining potential U_{a_R}. Then for every bounded continuous observable $\Phi : \mathbb{R}^{2d} \to \mathbb{R}$,*

$$\int_{\mathbb{R}^{2d}} \Phi(x - a_R, v)\,\pi_{N,1}^{a_R}(dx,dv) = \int_{\mathbb{R}^{2d}} \Phi(x,v)\,\pi_{N,1}^0(dx,dv)$$

for every N, and

$$\int_{\mathbb{R}^{2d}} \Phi(x - a_R, v)\,\pi_\infty^{a_R}(dx,dv) = \int_{\mathbb{R}^{2d}} \Phi(x,v)\,\pi_\infty^0(dx,dv)$$

for every R. Consequently the two limits commute:

$$\lim_{R\to\infty}\lim_{N\to\infty} \int \Phi(x-a_R,v)\,\pi_{N,1}^{a_R}(dx,dv) = \lim_{N\to\infty}\lim_{R\to\infty} \int \Phi(x-a_R,v)\,\pi_{N,1}^{a_R}(dx,dv) = \int \Phi(x,v)\,\pi_\infty^0(dx,dv).$$

Proof. The first identity is exactly the pushforward relation $(t_{-a_R})_\# \pi_{N,1}^{a_R} = \pi_{N,1}^0$, which follows from Theorem 8.11(iii) by taking first marginals. The second identity is $(t_{-a_R})_\# \pi_\infty^{a_R} = \pi_\infty^0$, which is Theorem 8.11(iv). The commutation of the $N \to \infty$ and $R \to \infty$ limits is then immediate, because after recentering the integral is independent of R, and the $N \to \infty$ limit is Theorem 8.7. □

9 Discussion

The preceding analysis separates three logically distinct ingredients.

1. The kinetic backbone supplies the hypocoercive entropy gap.
2. The row-normalized viscous coupling is an N-uniform, dissipative perturbation on relative velocities.
3. The velocity squashing map is a deterministic observation-level post-processing step that preserves all probability metrics relevant to the convergence theory.

This decomposition is exactly what makes the kinetic core easier than the full fractal gas. Once cloning and adaptive diffusion are removed, the only genuinely new interaction is the viscous velocity alignment, and its role is stabilizing rather than destabilizing.

Two extensions are intentionally left outside the scope of this note.

- The adaptive geometric extension, in which isotropic noise is replaced by state-dependent anisotropic diffusion.
- The full algorithmic operator, in which the kinetic stage is coupled to cloning, companion selection, and the fitness potential.

Those additions require a separate perturbative argument. The present paper is the clean kinetic backbone on which such extensions are built. At the numerical level, the continuous kinetic stage can be paired with splitting integrators for Langevin dynamics, while the smooth cap is applied only at observation times [3].

References

[1] Dominique Bakry and Michel Émery. Diffusions hypercontractives. In *Séminaire de Probabilités XIX 1983/84*, volume 1123 of *Lecture Notes in Mathematics*, pages 177–206. Springer, 1985.

[2] Persi Diaconis and David Freedman. Finite exchangeable sequences. *The Annals of Probability*, 8(4):745–764, 1980.

[3] Ben Leimkuhler and Charles Matthews. *Molecular Dynamics: With Deterministic and Stochastic Numerical Methods*, volume 39 of *Interdisciplinary Applied Mathematics*. Springer, 2015.

[4] Sean P. Meyn and Richard L. Tweedie. *Markov Chains and Stochastic Stability*. Cambridge University Press, 2nd edition, 2012.

[5] Gareth O. Roberts and Richard L. Tweedie. Exponential convergence of langevin distributions and their discrete approximations. *Bernoulli*, 2(4):341–363, 1996.

[6] Alain-Sol Sznitman. Topics in propagation of chaos. In *École d'Été de Probabilités de Saint-Flour XIX — 1989*, volume 1464 of *Lecture Notes in Mathematics*, pages 165–251. Springer, 1991.

[7] Cédric Villani. *Hypocoercivity*, volume 202 of *Memoirs of the American Mathematical Society*. American Mathematical Society, 2009.